



Physics in the Architecture: Fitting PDEs with Neural Fields

Akshay Kamath, Sam Thiele et al.

HZDR · 25.09.2026

Introductions & Motivation



- Group lead “Digital Earth” at the HIF
- Structural geology / volcanology (geometry & mechanics)
- Long interest in ML + big data + hyperspectral imaging

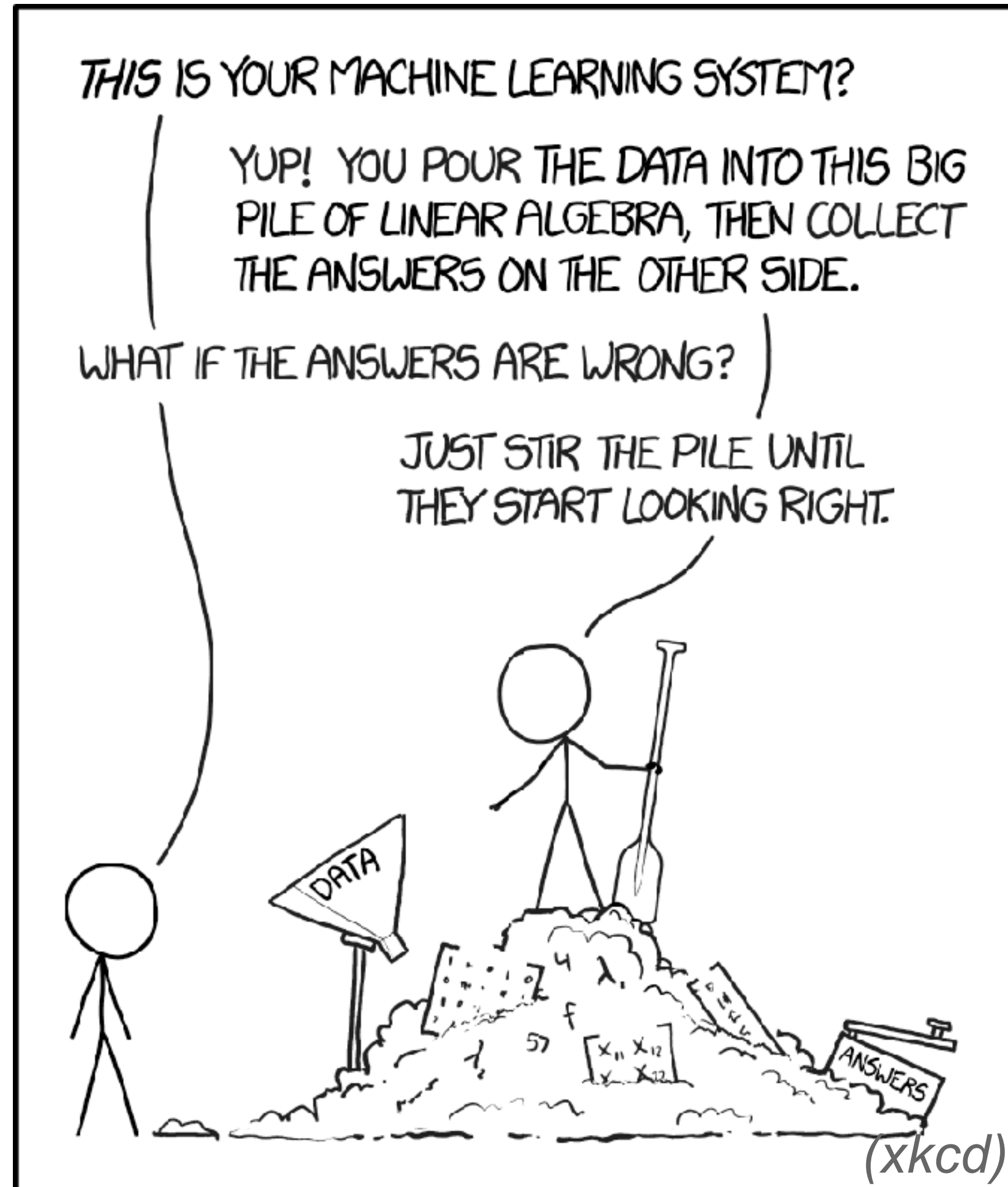
- PhD student at HIF —
- Geophysics background (gravity/magnetic fields) —
- Genuine math nerd —



Broad interests / topics:

- Coregistration and alignment of data
- Interpolation of sparse observations
- Neural representations of spatial $\pm$ temporal functions

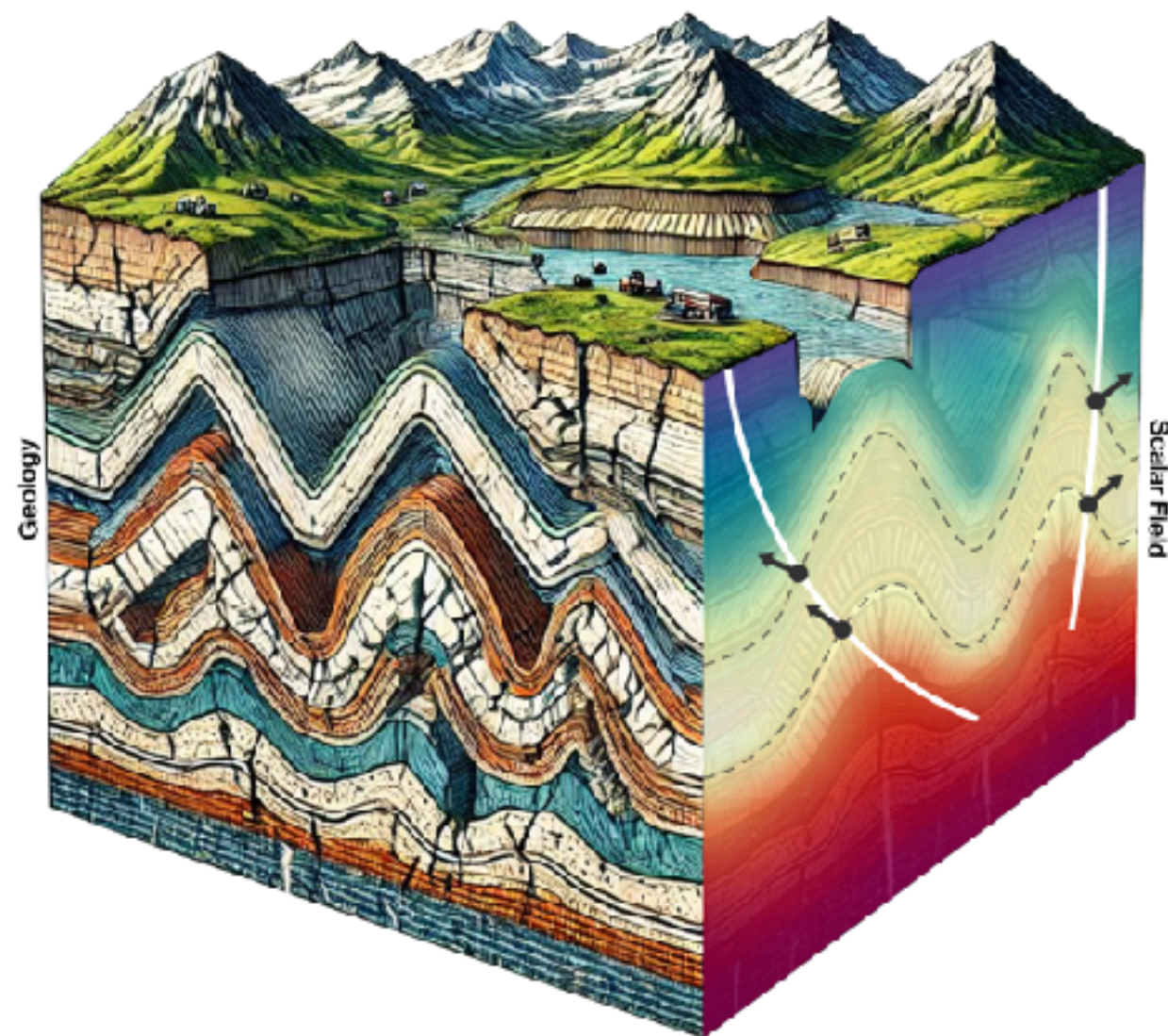
Workshop Outline & Aims



Aims & discussion points:

- Introduction to **neural fields, PINNs** and associated **challenges**
- Hands-on examples of **physically informed/constrained interpolation**
- *Input on different, new or better approaches :-)*
- *Other asks for which PINNs / related approaches might help?*

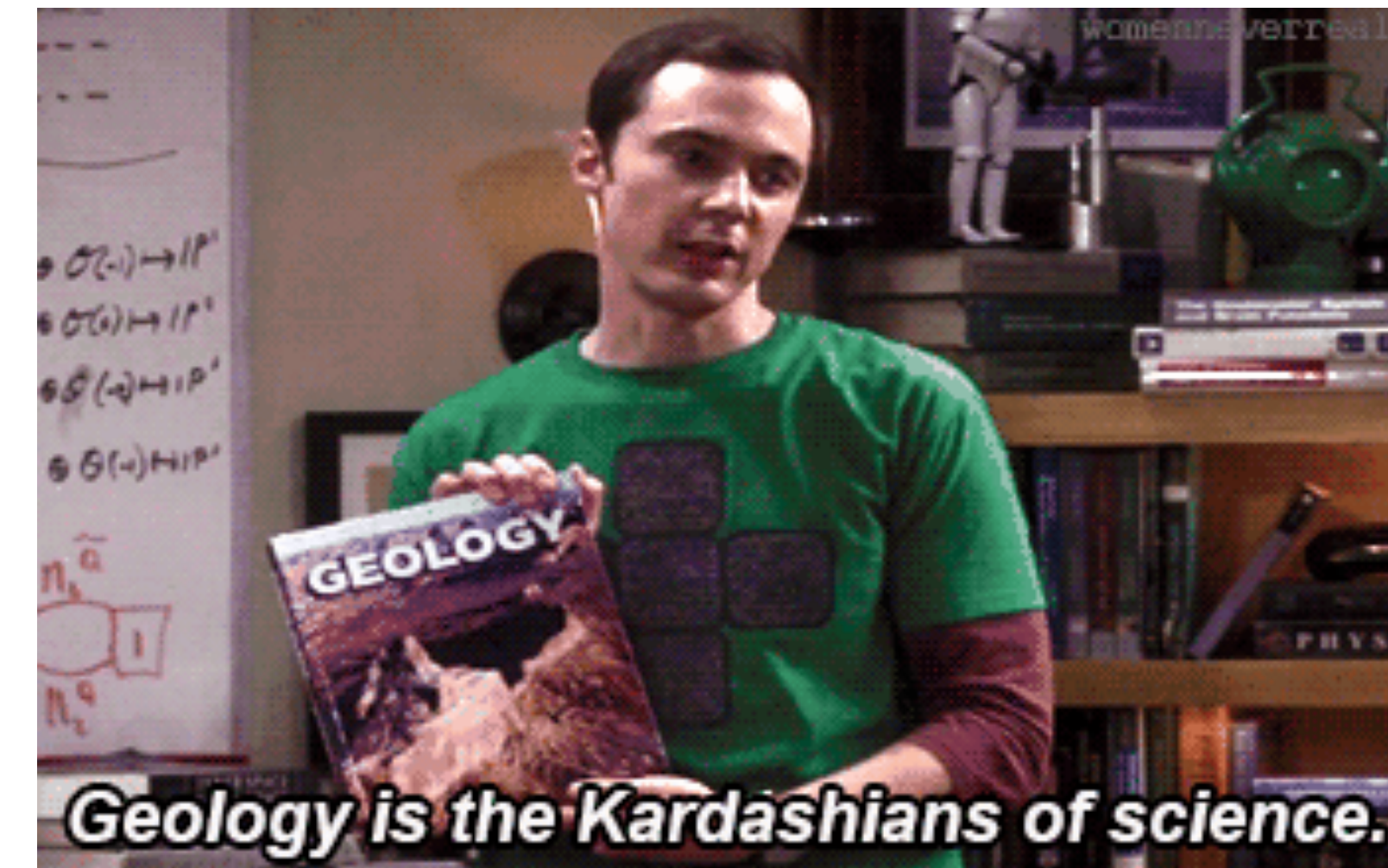
What's this got to do with rocks?



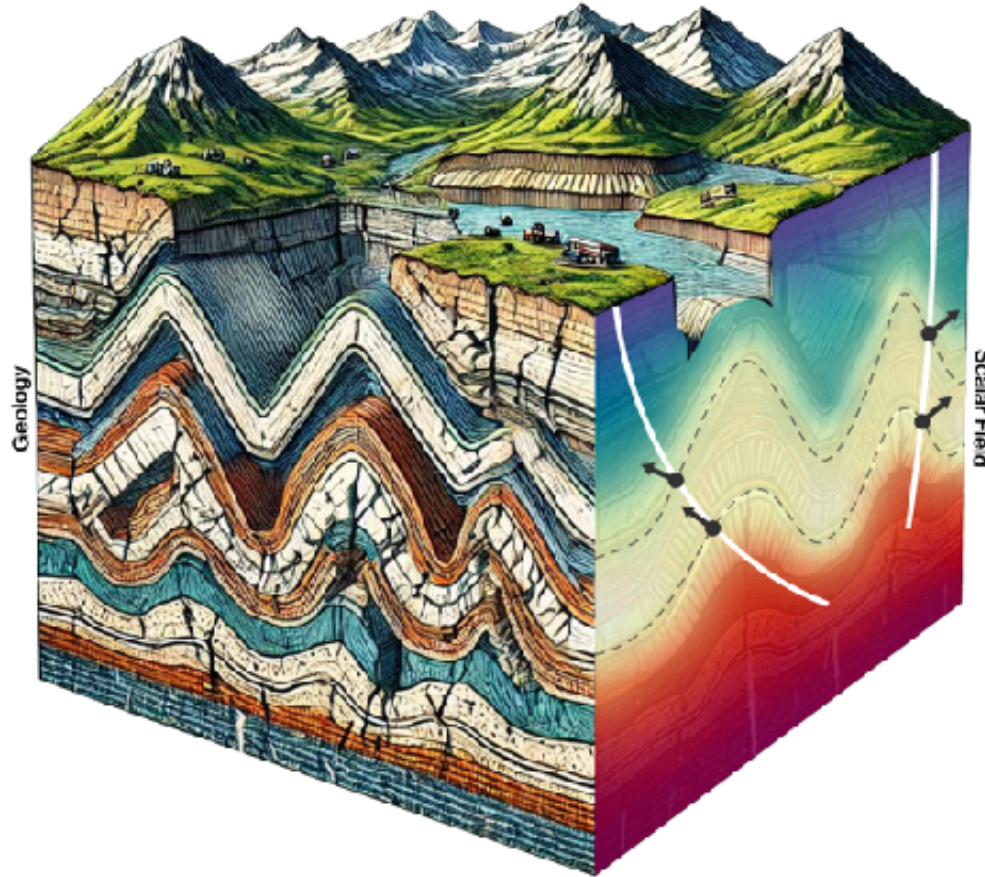
- Earth is built from (deformed) rock volumes
- Resulting geometry helps understand/predict:

1. **Earth history** (tectonics, mountain building, etc.)
2. **Subsurface flow** (water, nuclear waste, geothermal, hydrocarbons)
3. **Geohazards & engineering** (landslides, earthquakes, volcanoes, tunnels, ...)
4. **Mineral resources** (where's the gold, how do I get at it!)

- But rocks are very very hard to “see” through.
So how do we predict what's below our feet?



What's this got to do with rocks?

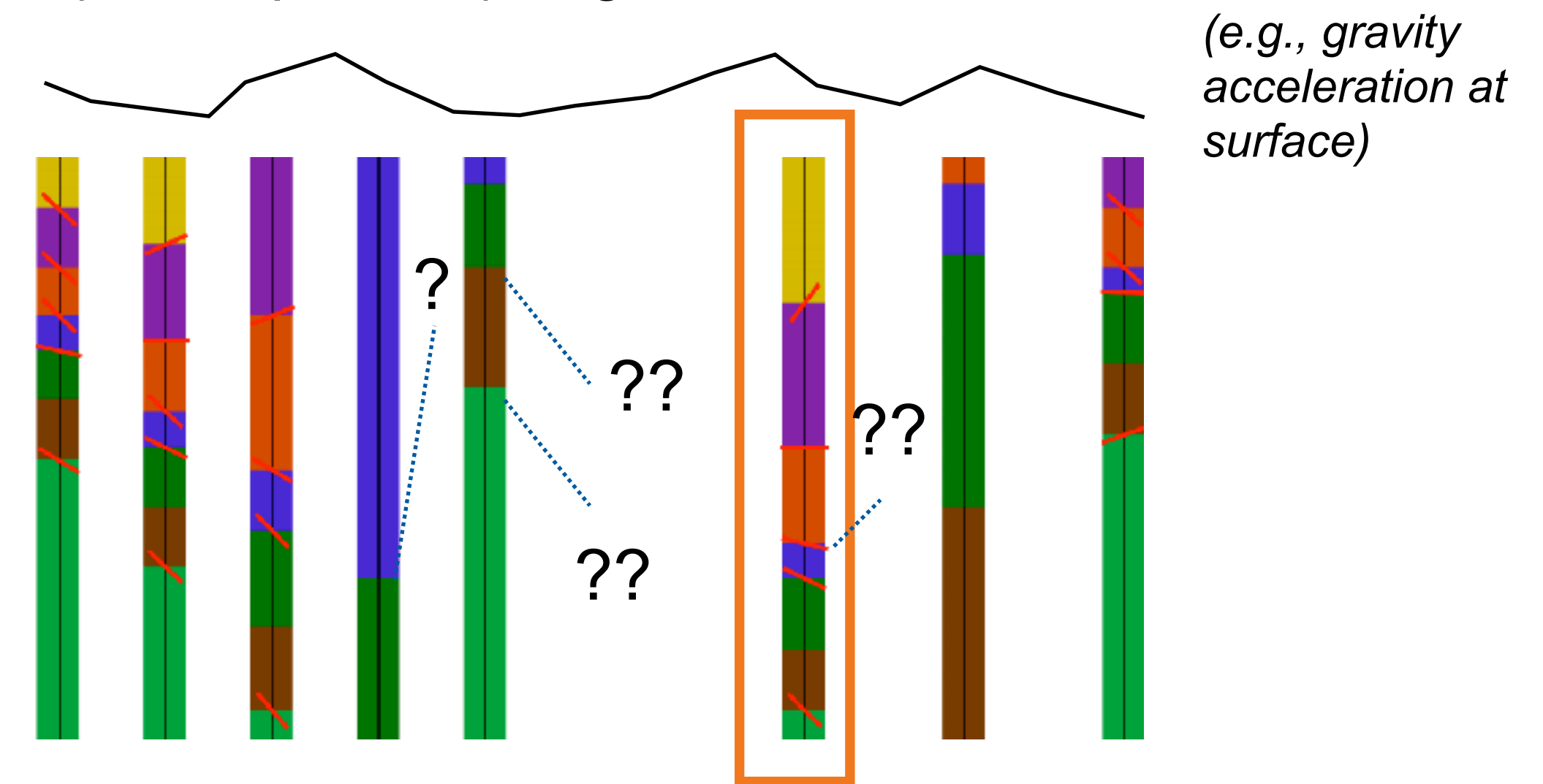


The real subsurface is a messy 3D world...

Challenges

- **Sparse** surface/subsurface data
- **Under-constrained** inverse problem (many solutions).
- **Disparate-constraints** - solution should satisfy many different data types and (geological) rules.
- **Bias** - commonly used data (e.g., geological logs) suffer from non-trivial interpretation bias

...But we observe only indirect signals (geophysics) and small (and expensive) fragments.

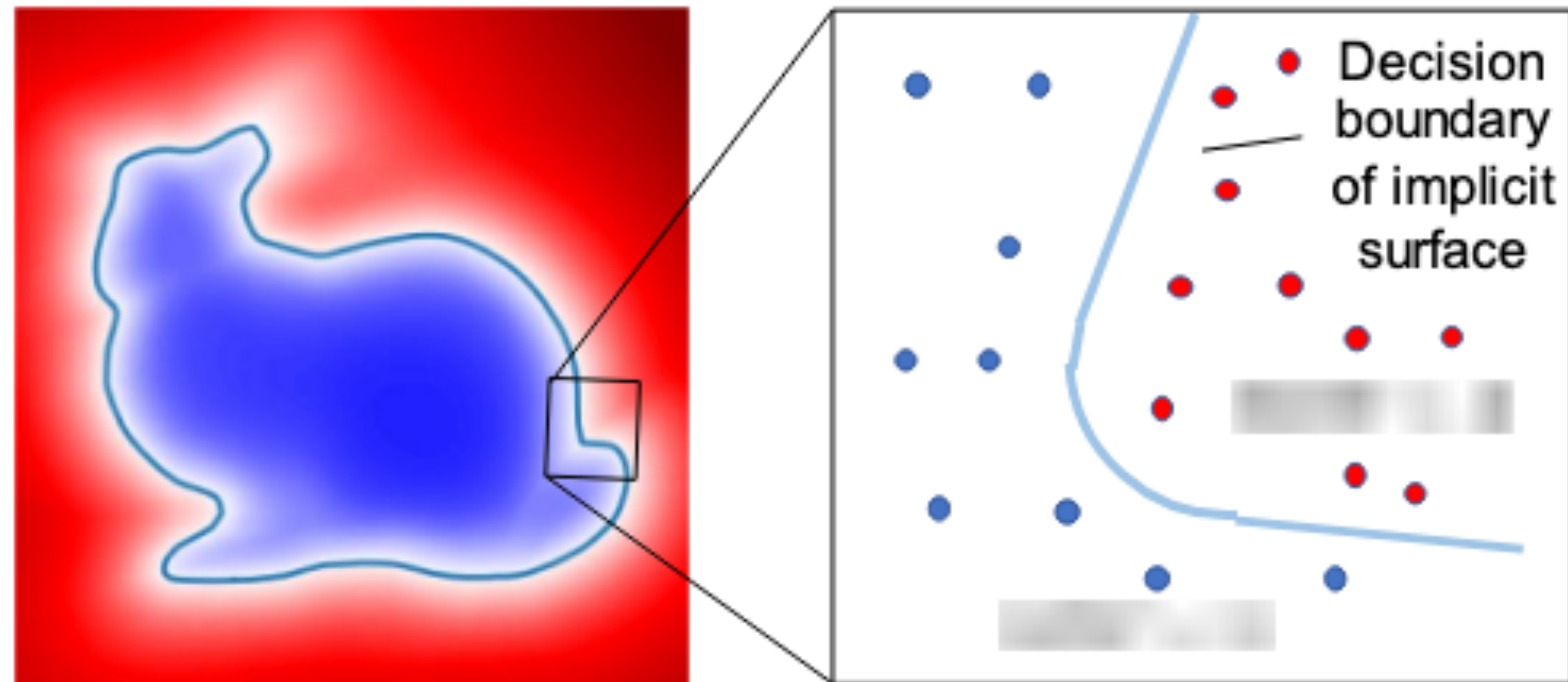


~0.1-1 million € per drillhole!

...and need to “guess” (interpolate) the rest.

What's this got to do with rocks?

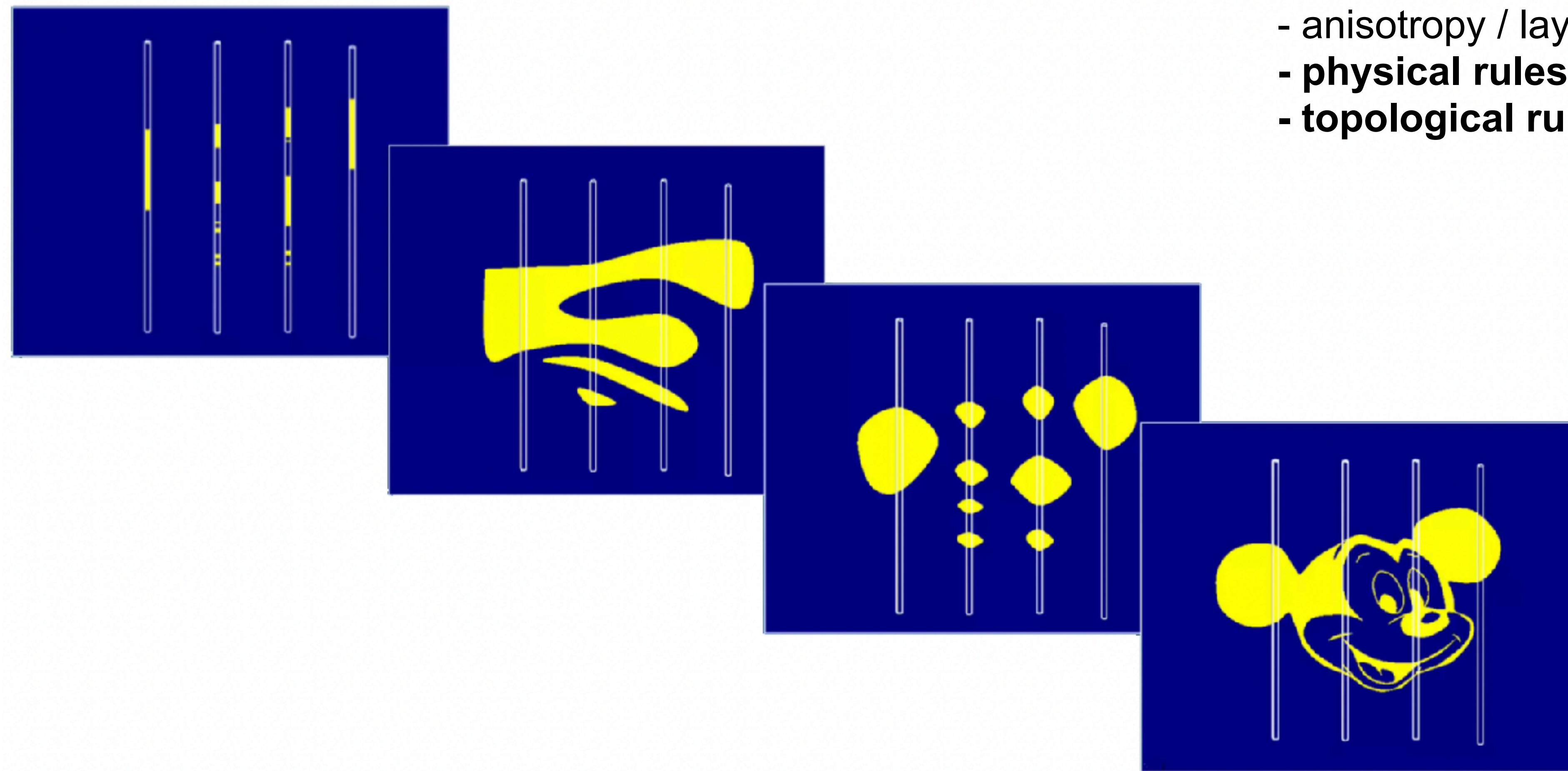
State-of-the-art approaches use implicit interpolation.



What's this got to do with rocks?

But... interpolation is (very) non-uniquemeaning we need constraints beyond data.

- smoothness?
- anisotropy / layering?
- **physical rules?**
- **topological rules?**

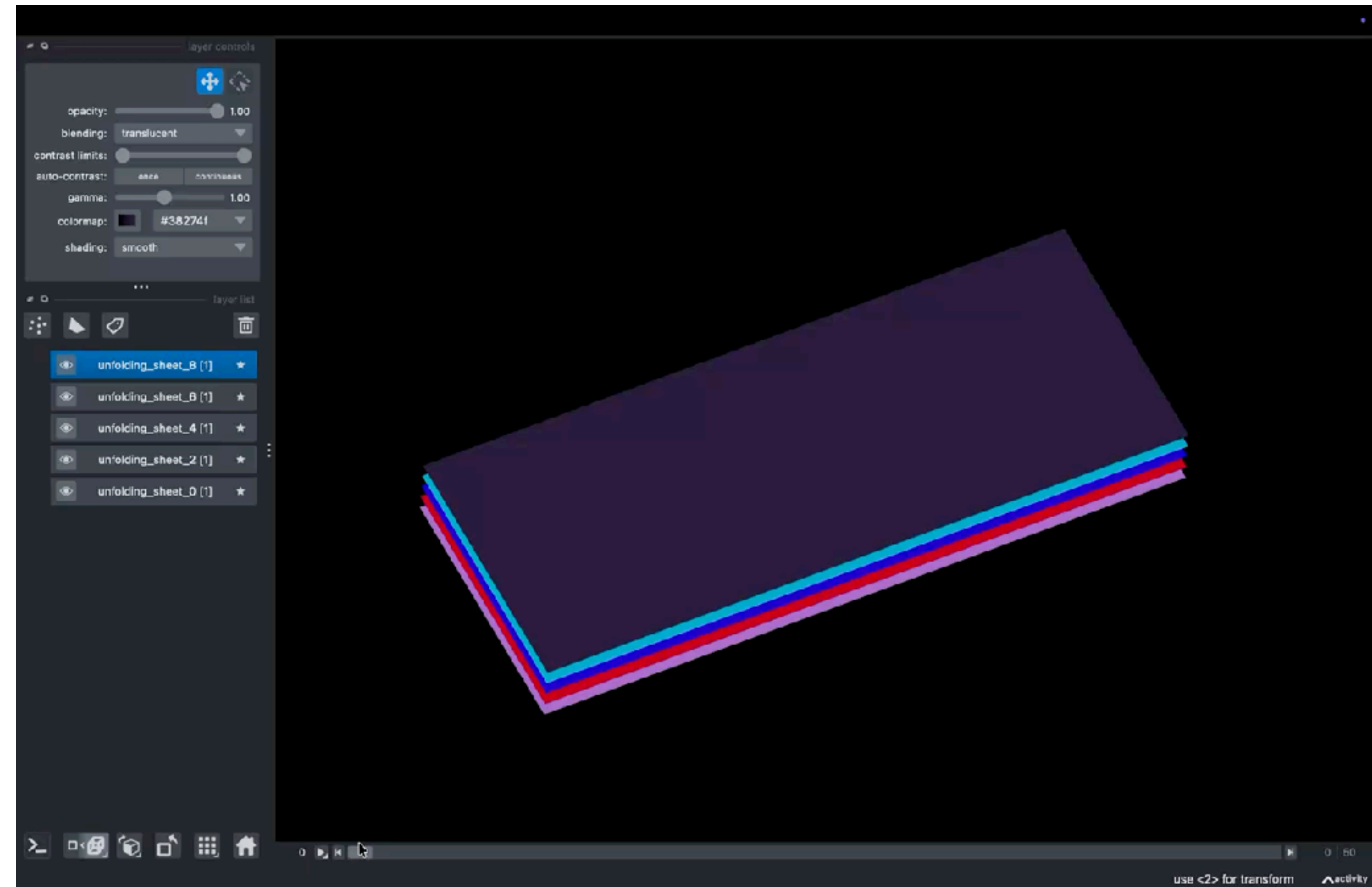


What's this got to do with rocks?

Neural fields are an excellent tool for constrained interpolation, integrating:

- **process** (physics)
- **rules** (topology)
- **disparate data** (mapping, drilling, geophysics)

To find a “common earth model” that plausibly explains all observations while respecting geological / physical rules.



Workshop Outline & Aims

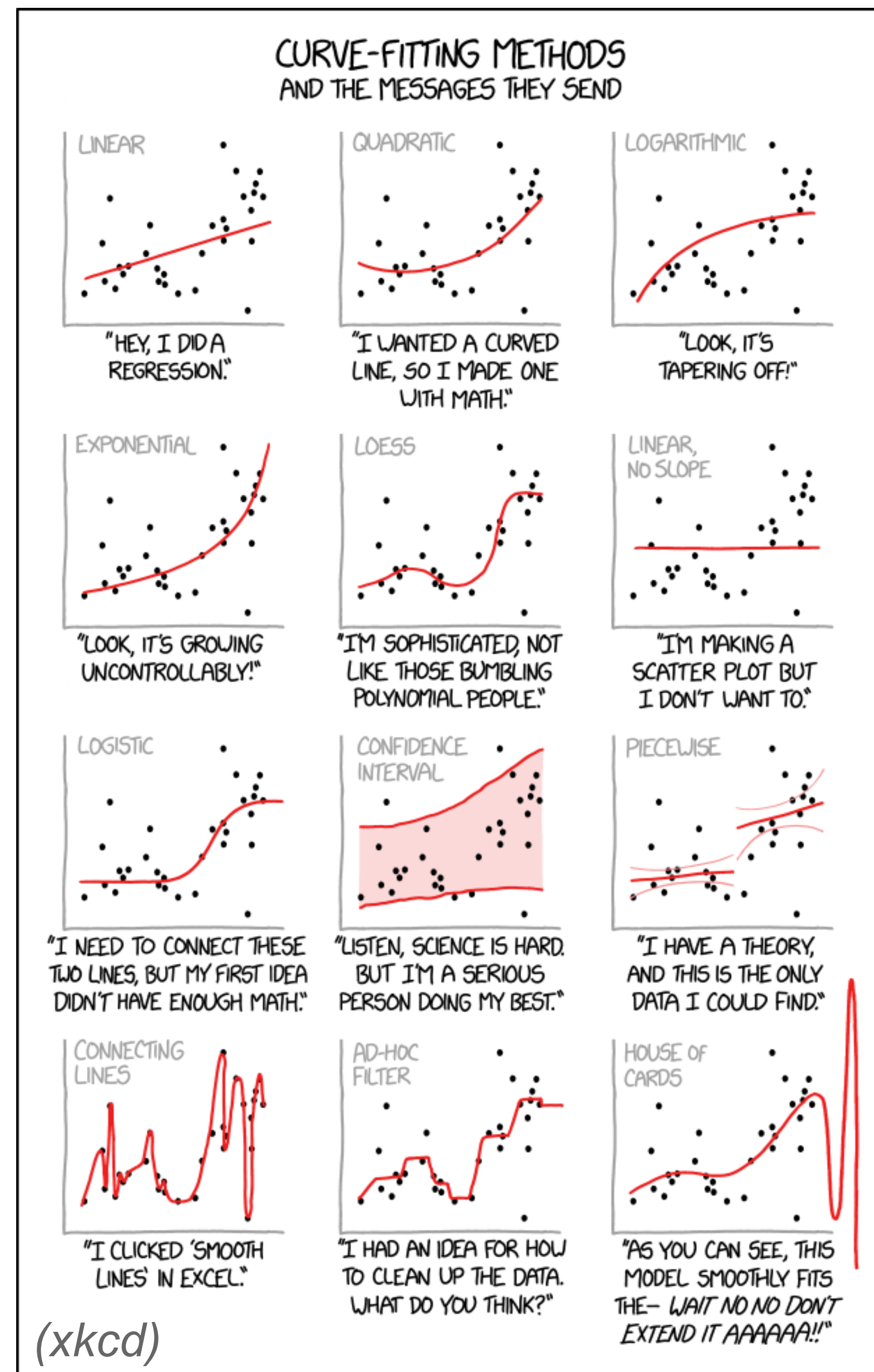
Part 1 - Theory / overview (30 to 45 minutes)

- **What are neural fields?**
- **The spectral bias problem** - and possible solutions
- **Physics in the loss** - PINNs and physically informed neural fields
- **Physics in the architecture** - avoiding hyper-parameter hell?

Part 2 - Interactive examples (45+ minutes)

- **Interpolating gravity fields** - from gravity gradiometry.
- **Learnable diffeomorphic interpolation** - undeforming data to a simple reference state.

Discussion and wrap-up (15 minutes)



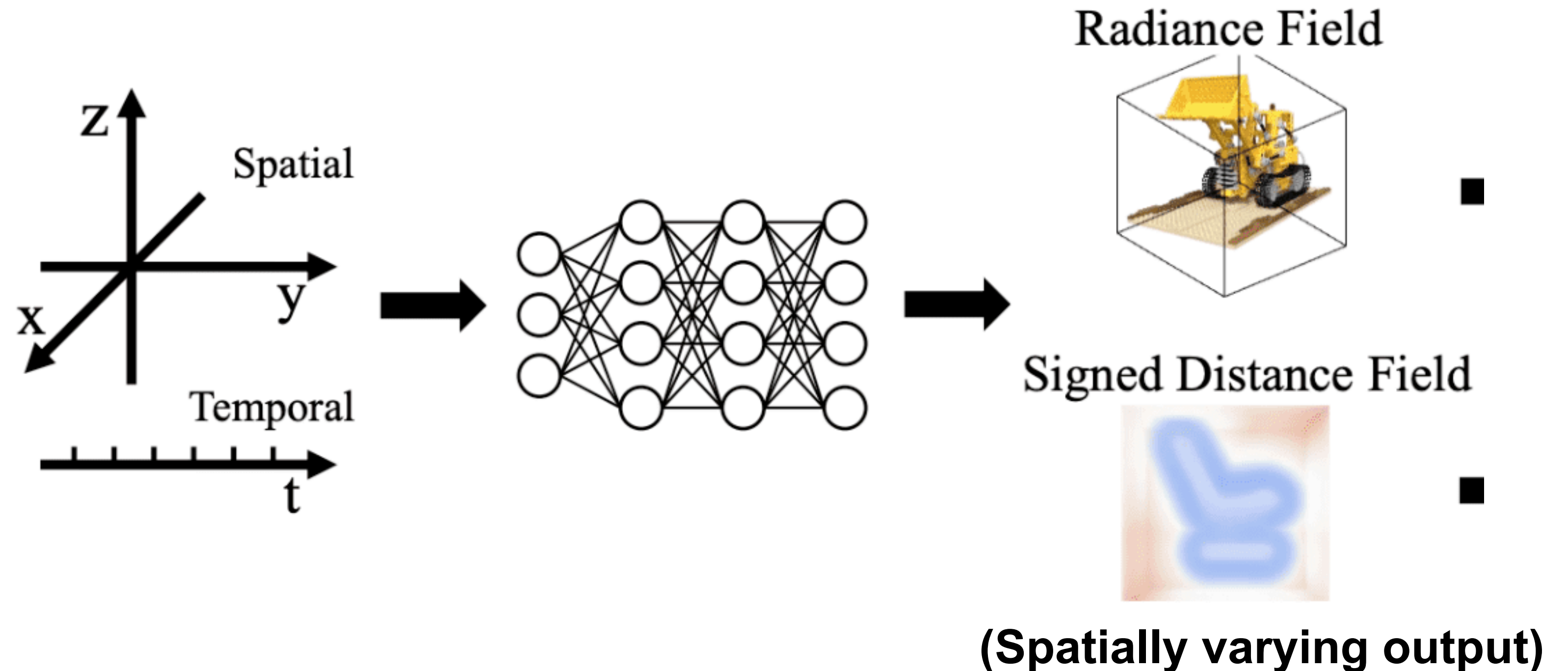
INRs - A new way to interpolate?

Neural Fields in Visual Computing and Beyond

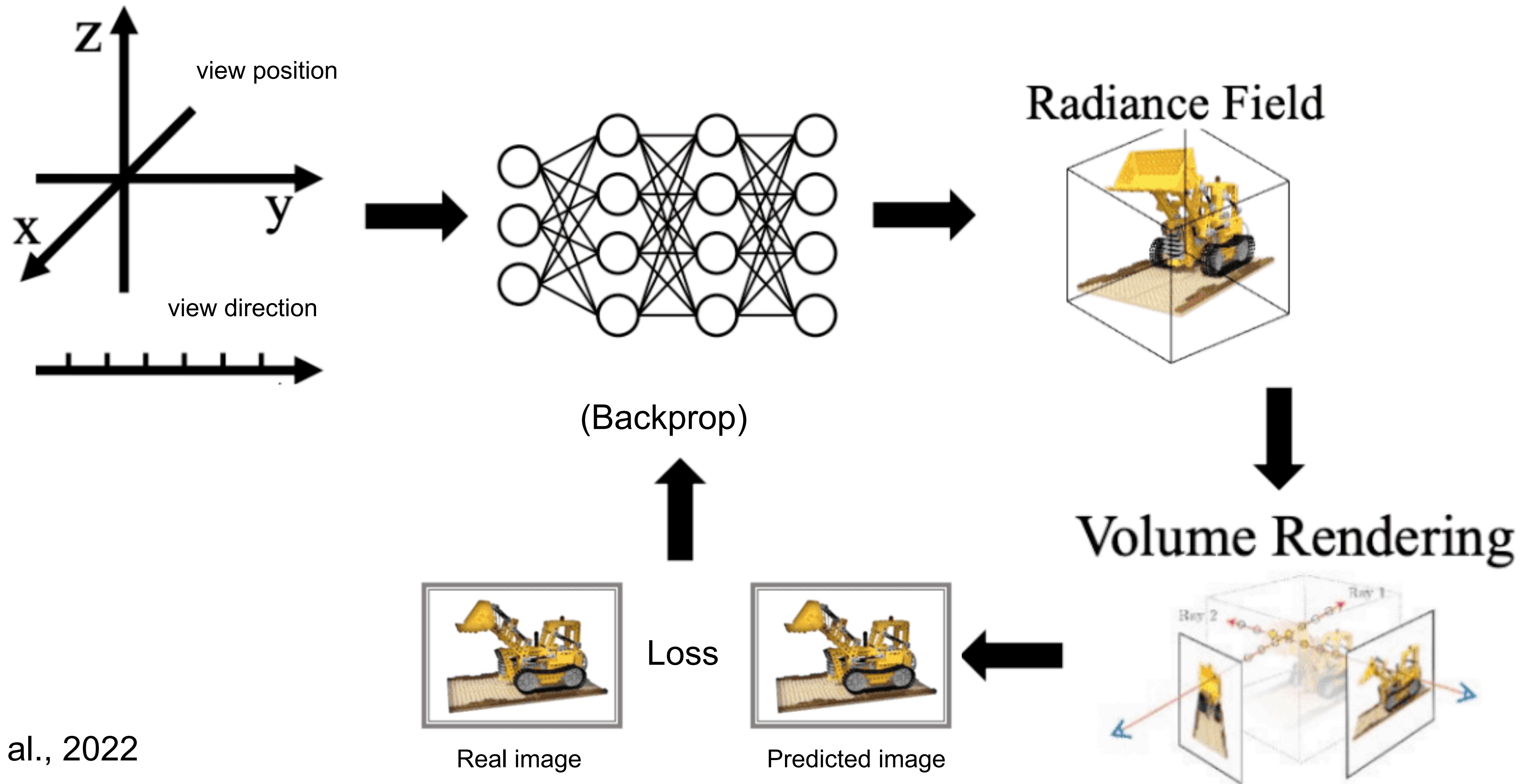
- INR $\rightarrow$ Spatial neural network. Input $x, y, z, \{t\}$ coordinates, output prediction.
- Can learn any “geometry” (universal approximation theorem).
- AutoDiff gives access to spatial (and if included, temporal) derivatives $\rightarrow$ **Can train on shapes instead of absolute values (Sobolev Training).**

Yiheng Xie^{1,2}  Towaki Takikawa^{3,4} Shunsuke Saito⁵  Or Litany⁴  Shiqin Yan¹ Numair Khan¹  Federico Tombari^{6,7}
James Tompkin¹  Vincent Sitzmann^{8†}  Srinath Sridhar^{1†} 

¹Brown University ²Unity Technologies ³University of Toronto ⁴NVIDIA ⁵Meta Reality Labs Research ⁶Google ⁷Technical University of Munich
⁸Massachusetts Institute of Technology [†]Equal advising



INRs - learning radiance fields (NeRFs)

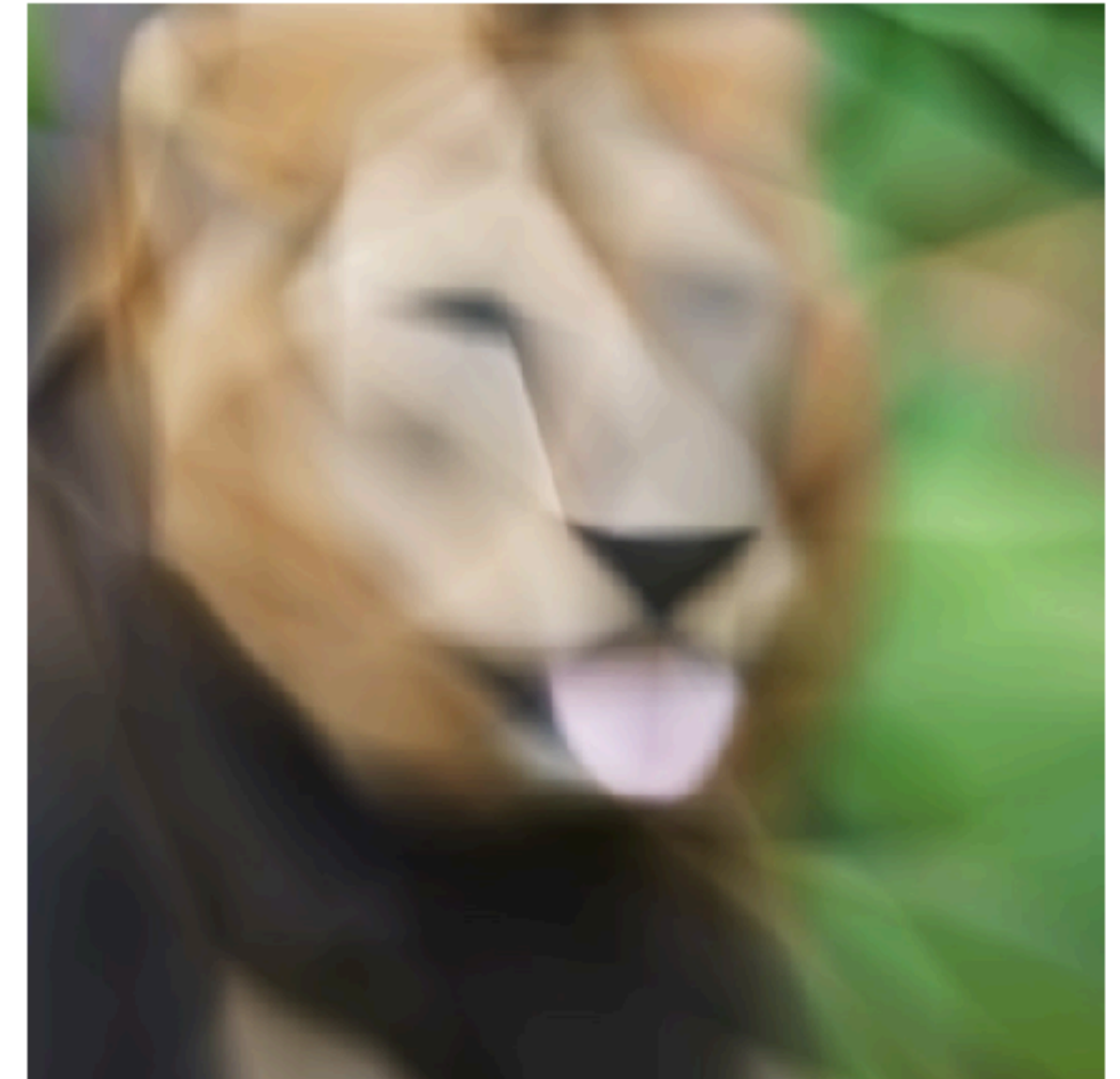
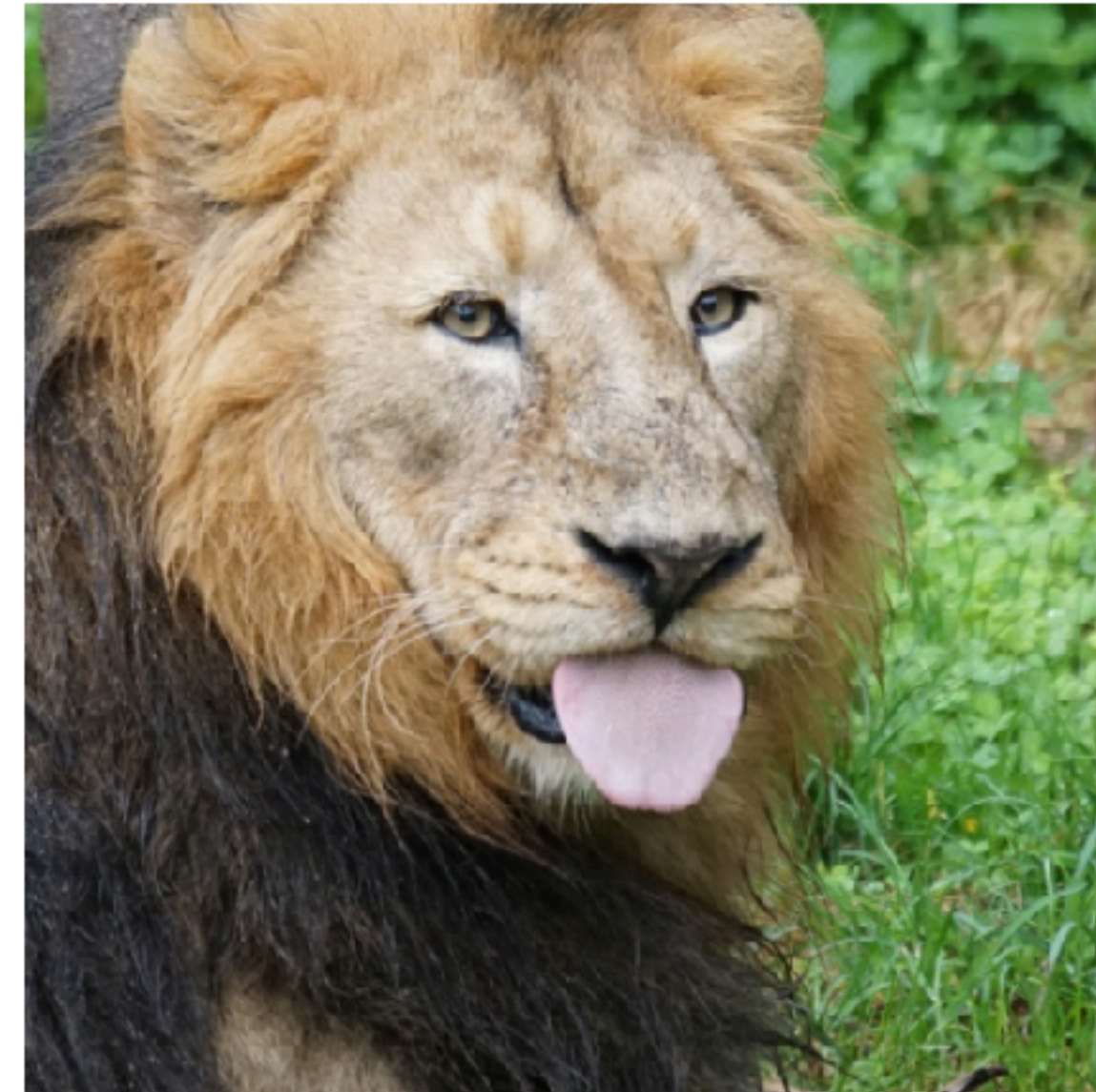


Xie et al., 2022

s.thiele@hzdr.de | a.kamath@hzdr.de

Easy? ...not so fast.

- MLPs tend to learn only low frequency features: they have a “**spectral bias**”.
- Low-dimensional input (2/3D) makes it difficult for the MLP to learn complex/intricate detail [*Tancik et al. (2020)*].



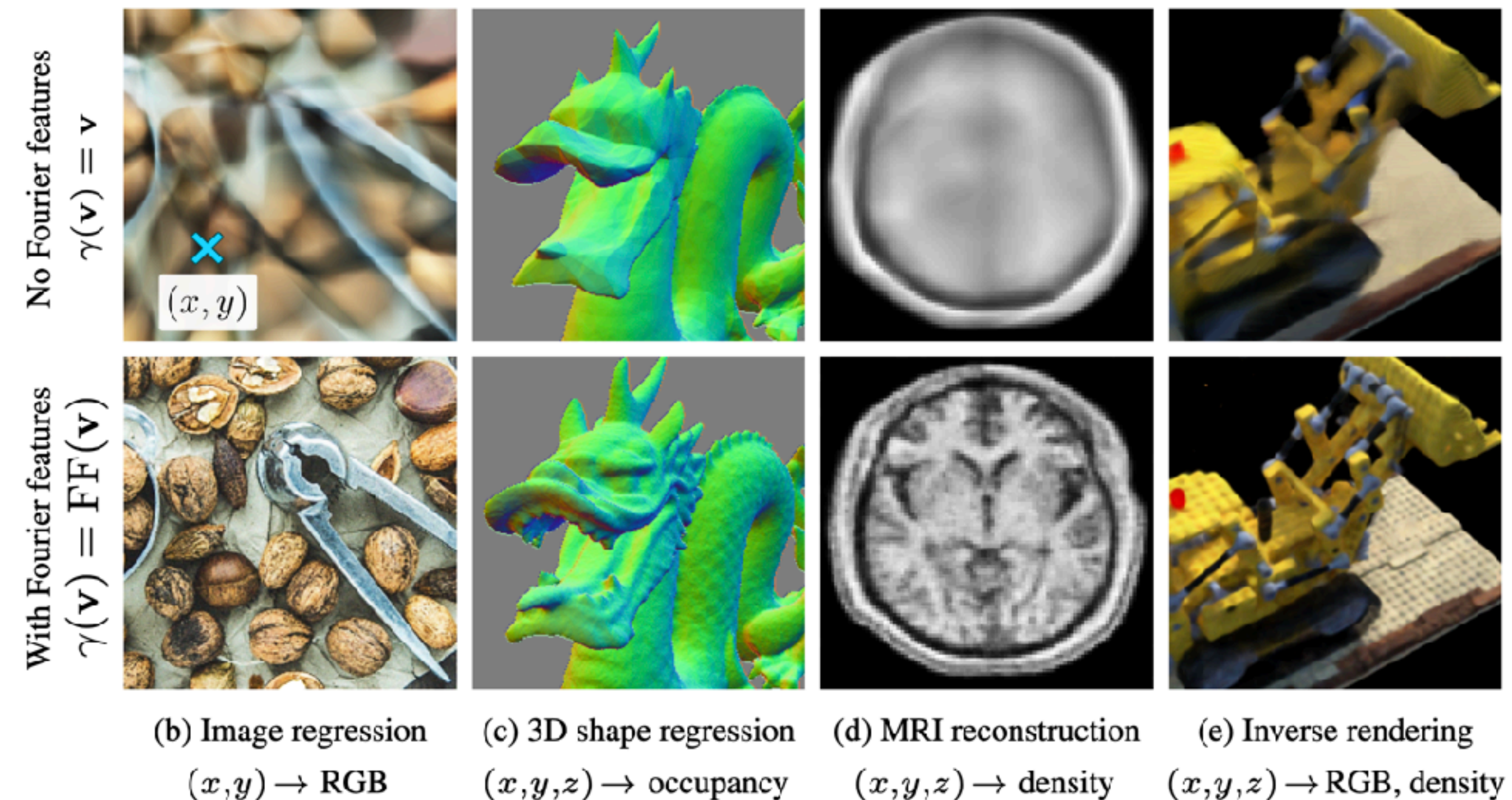
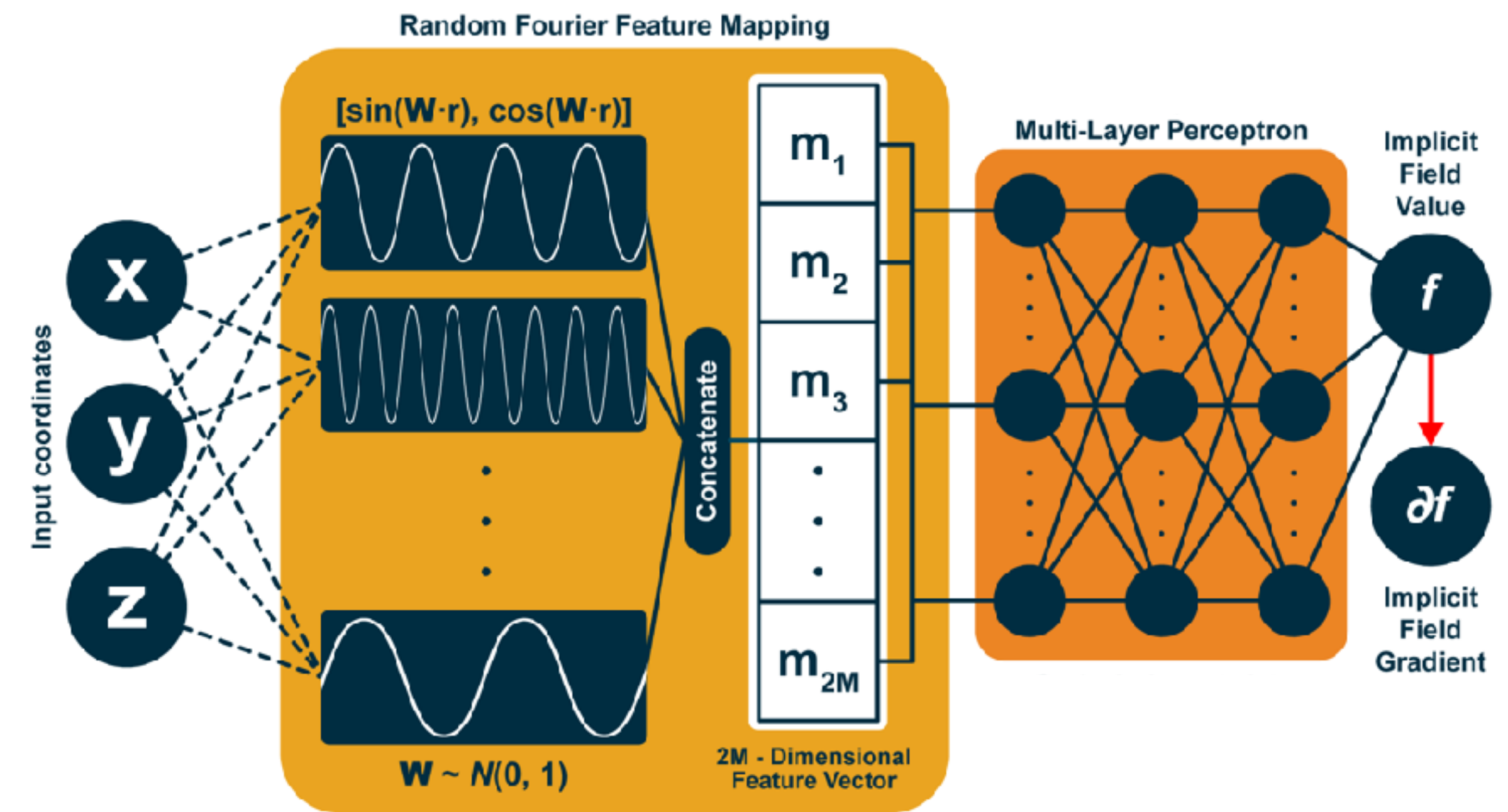
Ground Truth

Vanilla MLP

The solution: expand the dimensionality

Random Fourier Features (RFF)

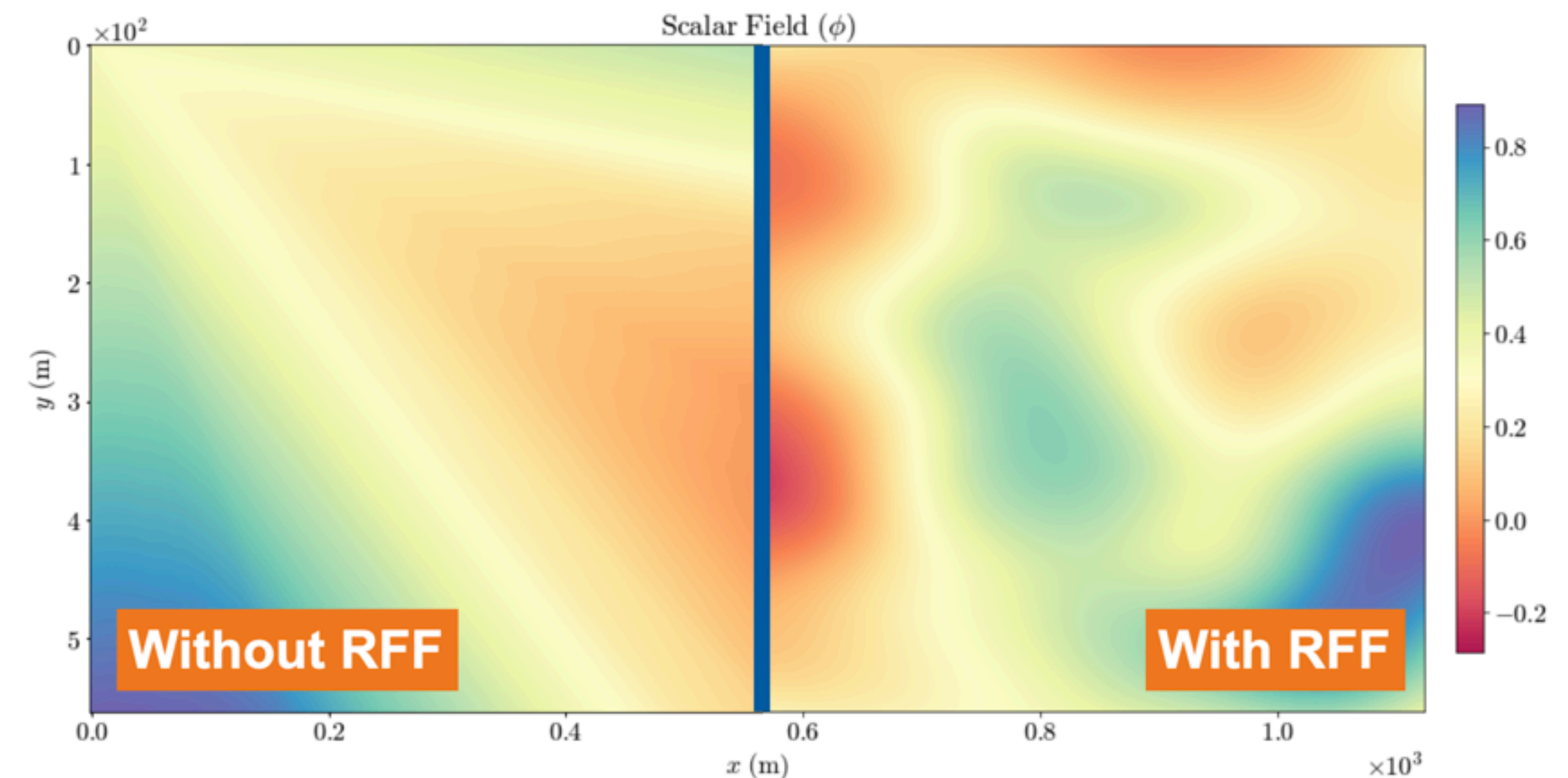
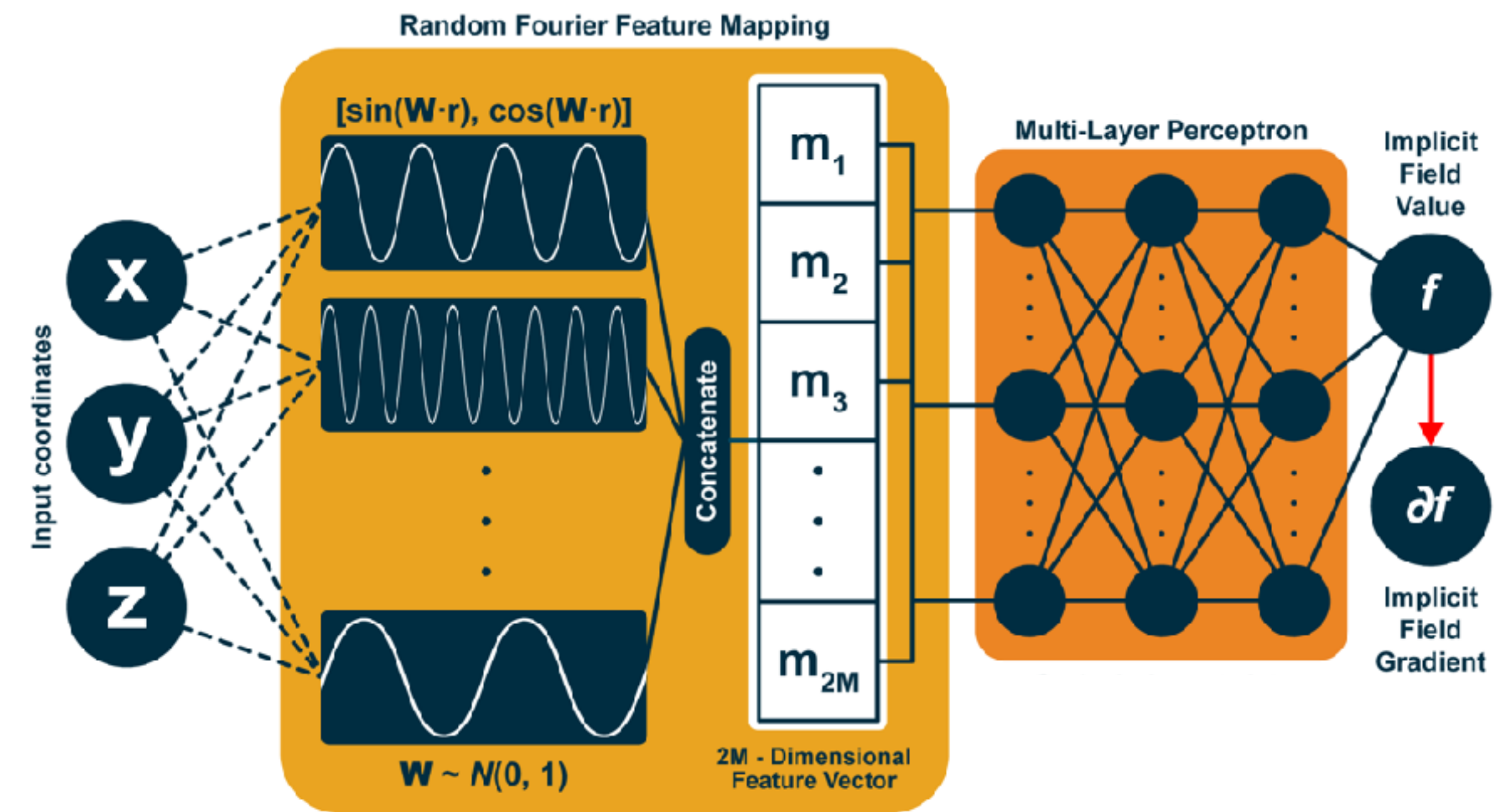
- $\vec{v} = [\cos(2\pi\mathbf{B}\vec{r}), \sin(2\pi\mathbf{B}\vec{r})]$
- $\mathbf{B} \sim \mathcal{N}(0, 1)$, Controlled by length scales.
- Allows network to learn intricate patterns within low-dimensional signals.
- Acts as a combination of positional encoding and sinusoidal activation function.
- Speeds up convergence significantly.
- (Conceptualized as kernel approximations by [Rahimi and Recht \(2007\)](#), implemented for neural networks in [Tancik et al., \(2020\)](#)).



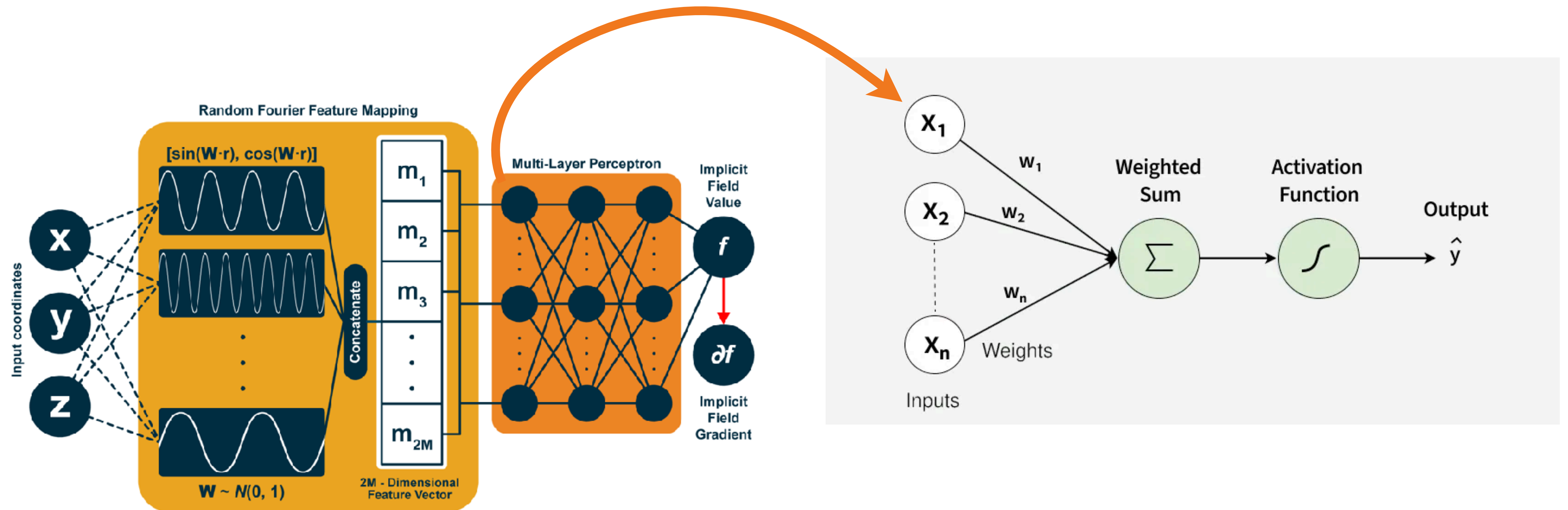
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Sidenote: Fourier Series Representations - Simple and analytically differentiable



$$s_N(x) = a_0 + \sum_{n=-\infty}^{\infty} \left(a_n \cos \left(2\pi \frac{n}{P} x \right) + b_n \sin \left(2\pi \frac{n}{P} x \right) \right)$$

Equivalent to a Pseudo-Fourier Series - The frequencies are drawn randomly!

The solution II: periodic activation functions (SIREN)

SIREN activation functions

- Move the periodic component (sinusoid) into the activation function (of last, or sometimes all, layers)
- Similar effect to RFFs
- But; requires careful initialization strategies (and scaling of input coordinates)

Implicit Neural Representations with Periodic Activation Functions

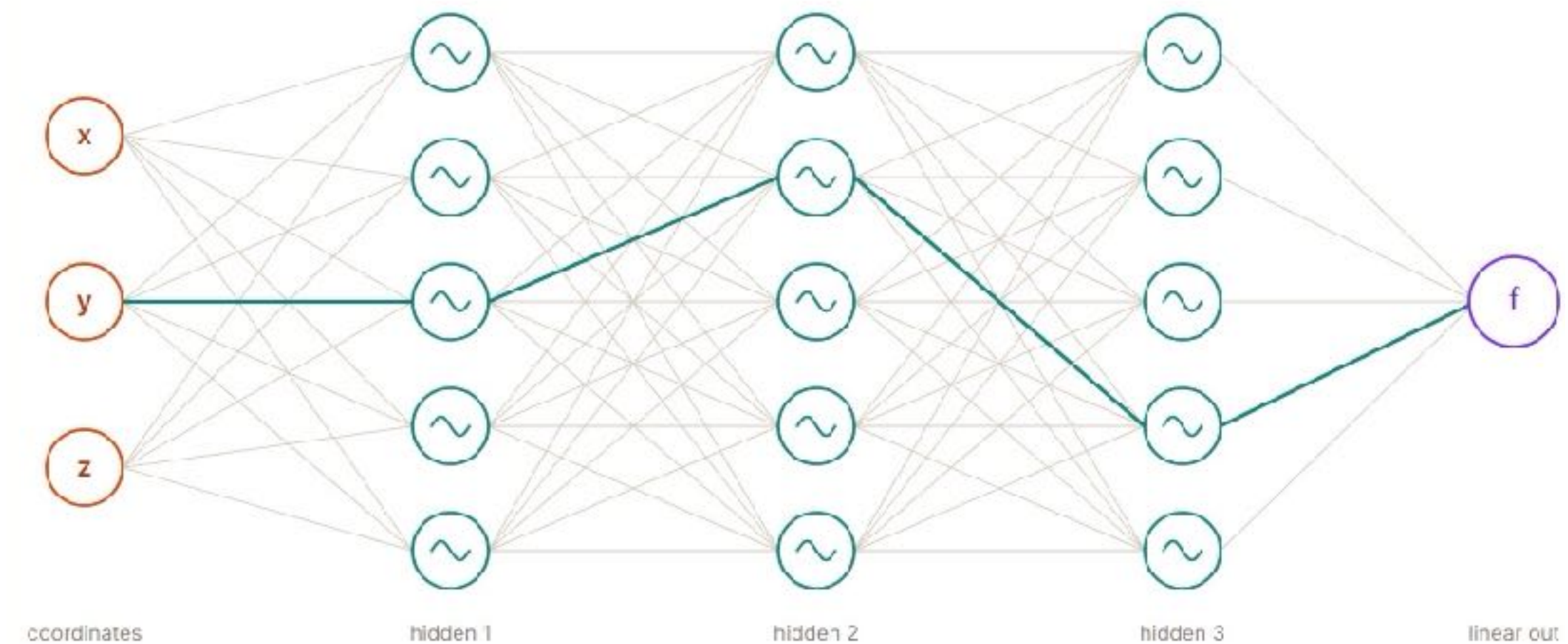
Vincent Sitzmann*
sitzmann@cs.stanford.edu

Julien N. P. Martel*
jnmartel@stanford.edu

Alexander W. Bergman
awb@stanford.edu

David B. Lindell
lindell@stanford.edu

Gordon Wetzstein
gordon.wetzstein@stanford.edu



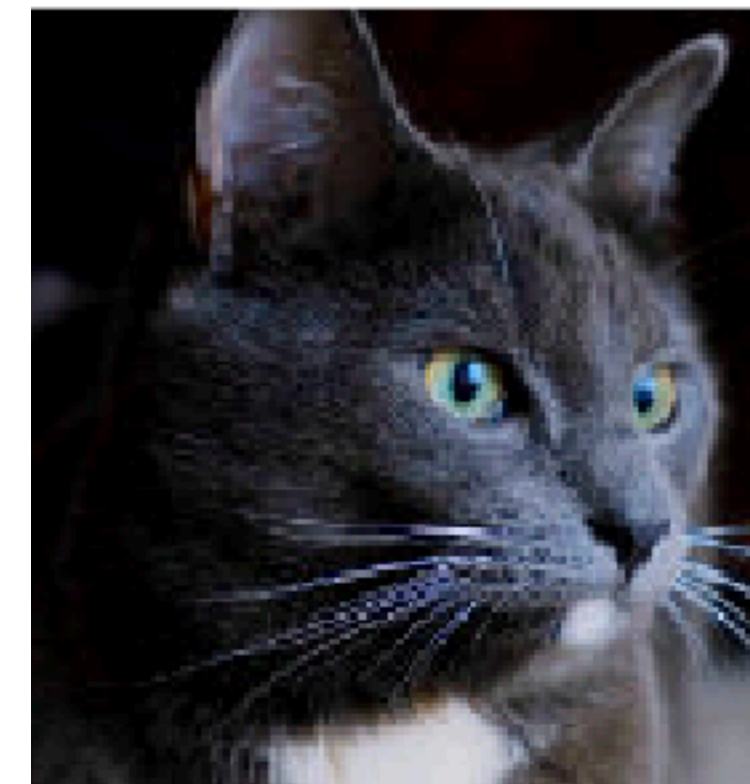
ReLU



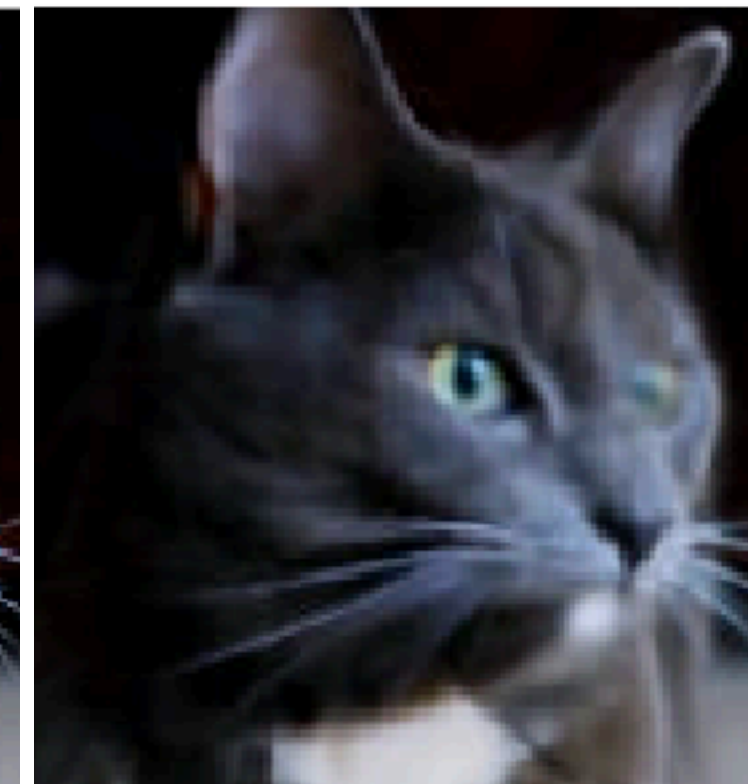
SIREN



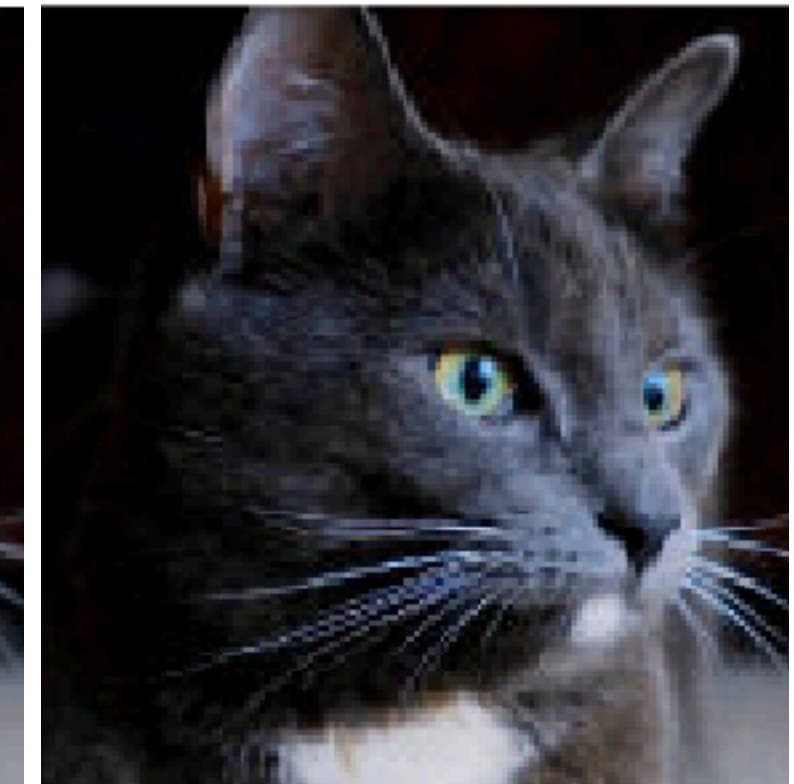
Ground Truth



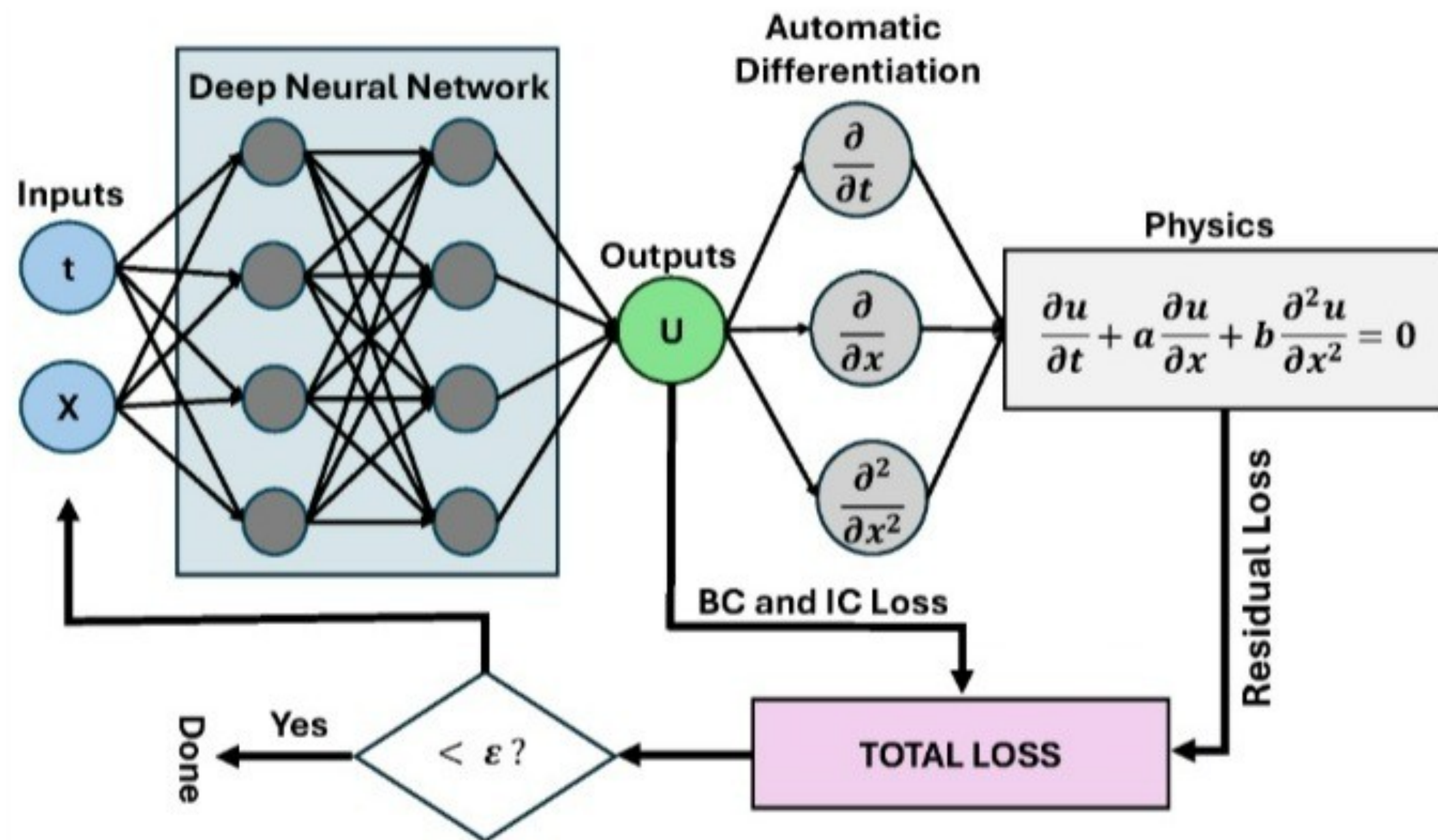
ReLU



SIREN



Physics-Informed Neural Networks (PINNs)



DGM: A deep learning algorithm for solving partial differential equations ☆,☆☆

Justin Sirignano^{a,*}, Konstantinos Spiliopoulos^b

Physics-Informed Neural Networks (PINNs) for Heat Transfer Problems

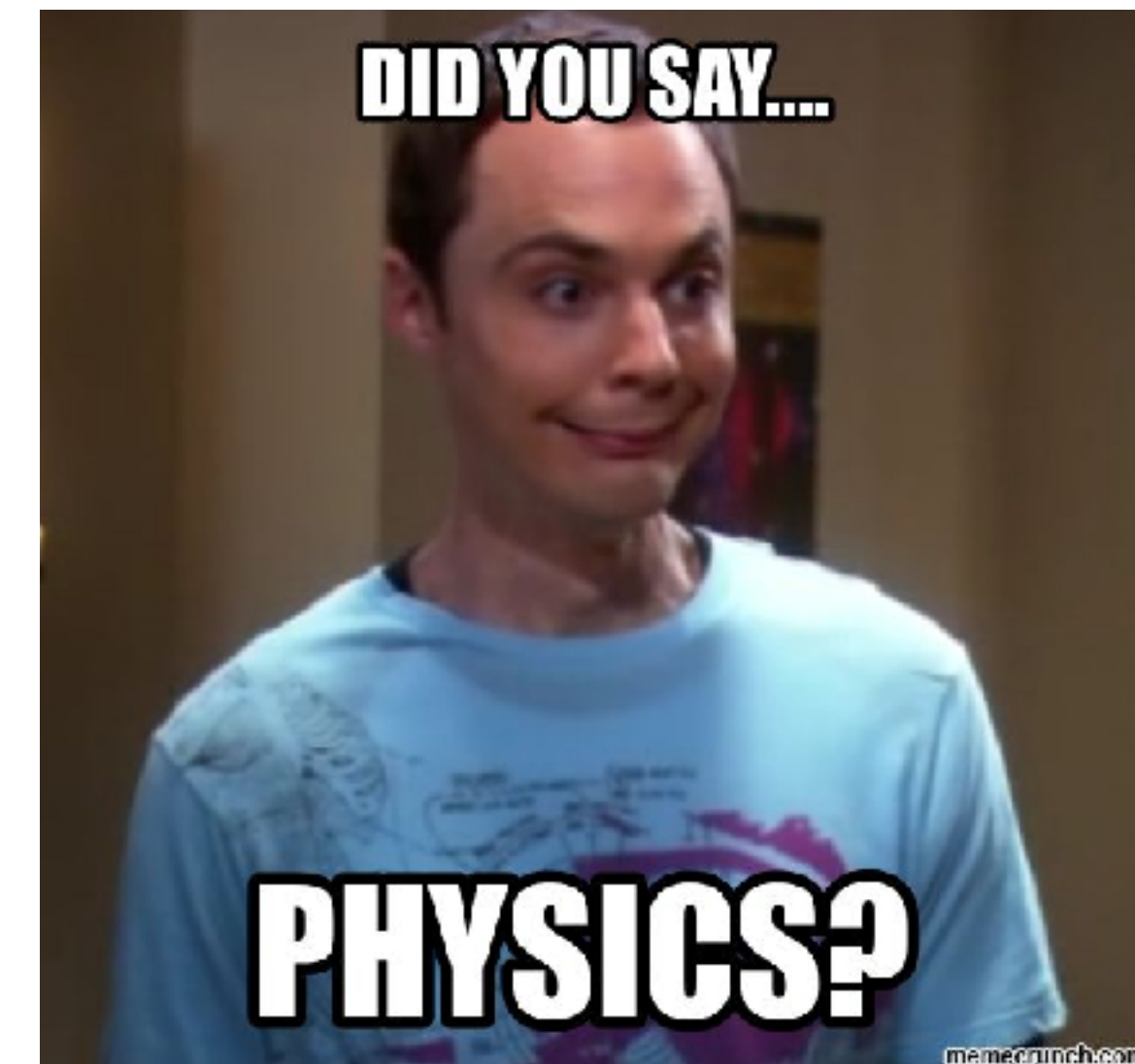
Shengze Cai*
Division of Applied Mathematics
Brown University
Providence, Rhode Island 02912
Email: shengze_cai@brown.edu

Zhicheng Wang*
Division of Applied Mathematics
Brown University
Providence, Rhode Island 02912
Email: zhicheng_wang@brown.edu

Sifan Wang
Graduate Group in Applied
Mathematics and Computational Science
University of Pennsylvania
Philadelphia, Pennsylvania 19104
Email: sifanw@sas.upenn.edu

Paris Perdikaris†
Department of Mechanical Engineering
and Applied Mechanics
University of Pennsylvania
Philadelphia, Pennsylvania 19104
Email: pgp@seas.upenn.edu

George Em Karniadakis‡
Division of Applied Mathematics and School of Engineering
Brown University
Providence, Rhode Island 02912
Email: george_karniadakis@brown.edu



Physics-informed neural networks (PINNs) for fluid mechanics: A review

Shengze Cai¹ · Zhiping Mao² · Zhicheng Wang³ · Minglang Yin^{4,5}
George Em Karniadakis^{1,4*}

A review on physics-informed machine learning for monitoring metal additive manufacturing process

Shoulun Yang, Shitong Peng*, Jianan Guo, and Fengtao Wang*

s.thiele@hzdr.de | a.kamath@hzdr.de

Applications of Physics-Informed Neural Networks in Power Systems - A Review

Bin Huang^{id}, Student Member, IEEE, and Jianhui Wang^{id}, Fellow, IEEE

Physics-Informed Neural Networks and Extensions

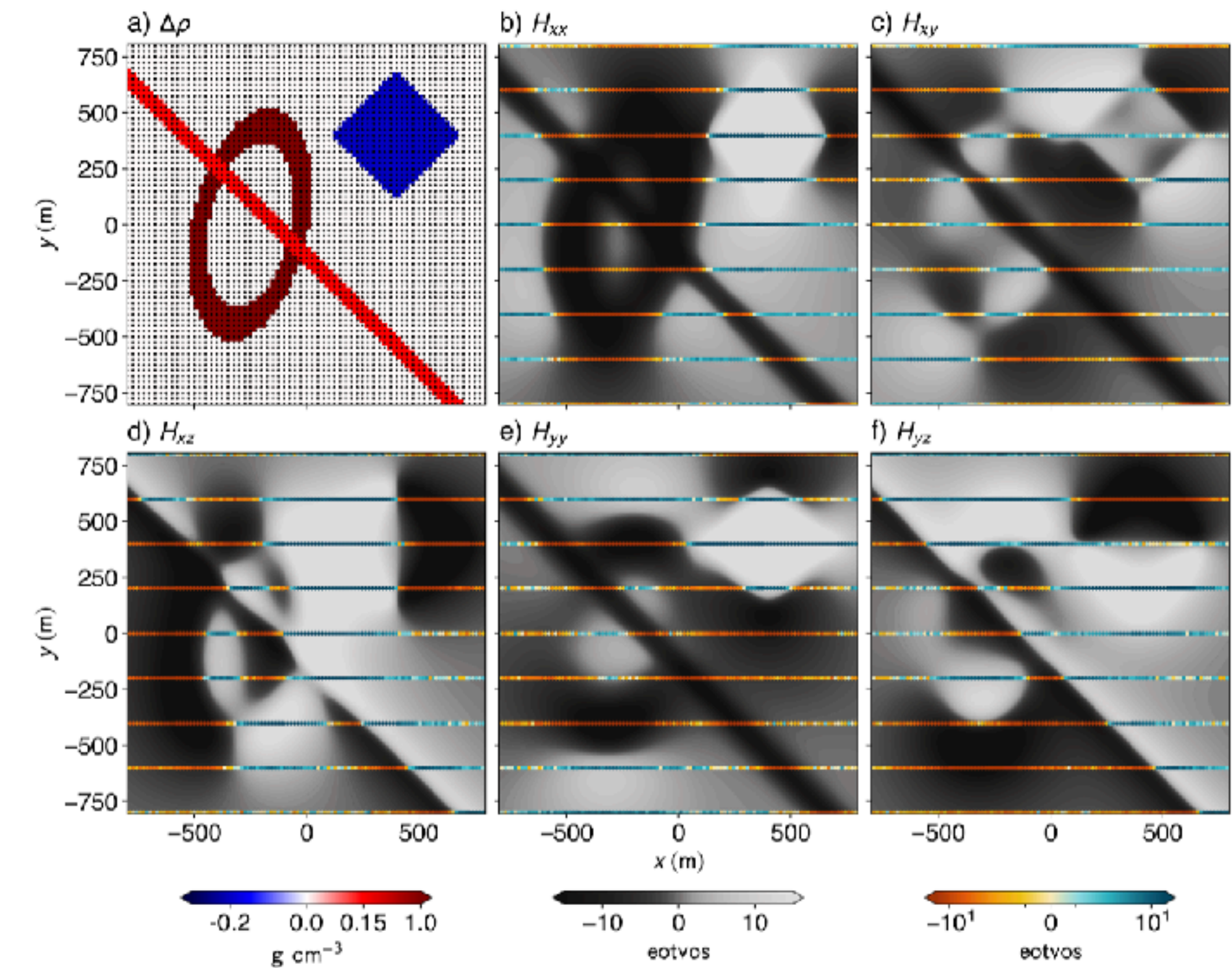
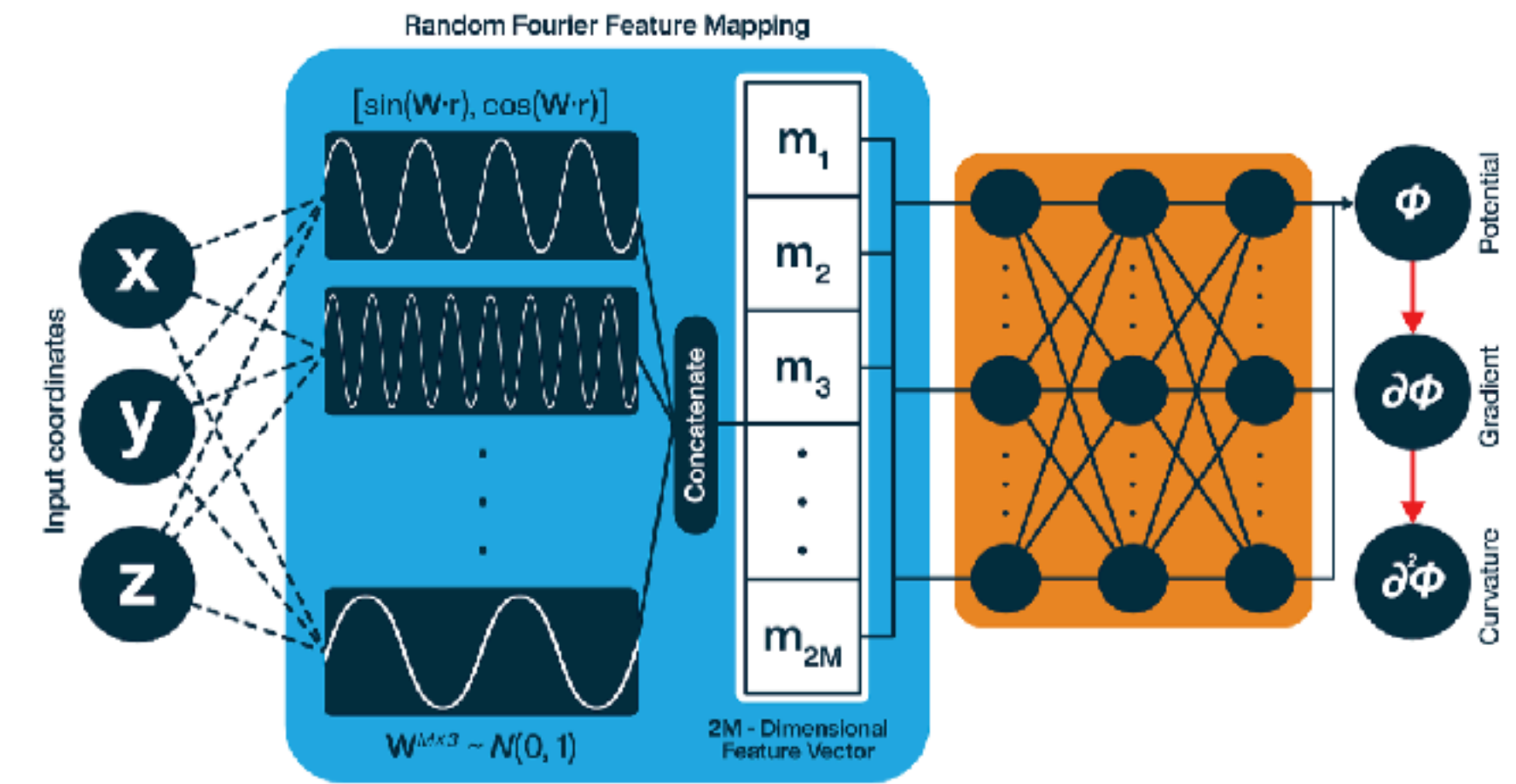
Maziar Raissi¹, Paris Perdikaris², Nazanin Ahmadi³, and George Em Karniadakis⁴

Scientific Machine Learning through Physics-Informed Neural Networks: Where we are and What's next

Salvatore Cuomo¹, Vincenzo Schiano Di Cola^{2*}, Fabio Giampaolo¹, Gianluigi Rozza², Maziar Raissi³ and Francesco Piccialli^{1*}

PINNs for Potential Field Gradiometry

- Gravity and Magnetic fields are measured to map subsurface contrasts (in density and susceptibility).
- Measured along flight-lines (tight in-line spacing, sparse cross-line).
- Tensor gradiometry measures components of the Hessian.
- Interpolation needs to be physically consistent (Laplace's eqn.)
- Solution: Make a NN predict a scalar field, get curvature with AD, fit to data along with a Laplacian loss.



$$\mathbf{v} = \nabla \phi \leftrightarrow \nabla \times \mathbf{v} = 0$$

$$\mathbf{H} = \nabla \mathbf{v} = \nabla(\nabla \phi) = \frac{\partial^2 \phi}{\partial r_i \partial r_j} \equiv \partial_i \partial_j \phi$$

$$\nabla^2 \phi \equiv \text{tr}(\mathbf{H}) = 0.$$

PINNs for Potential Field Gradiometry

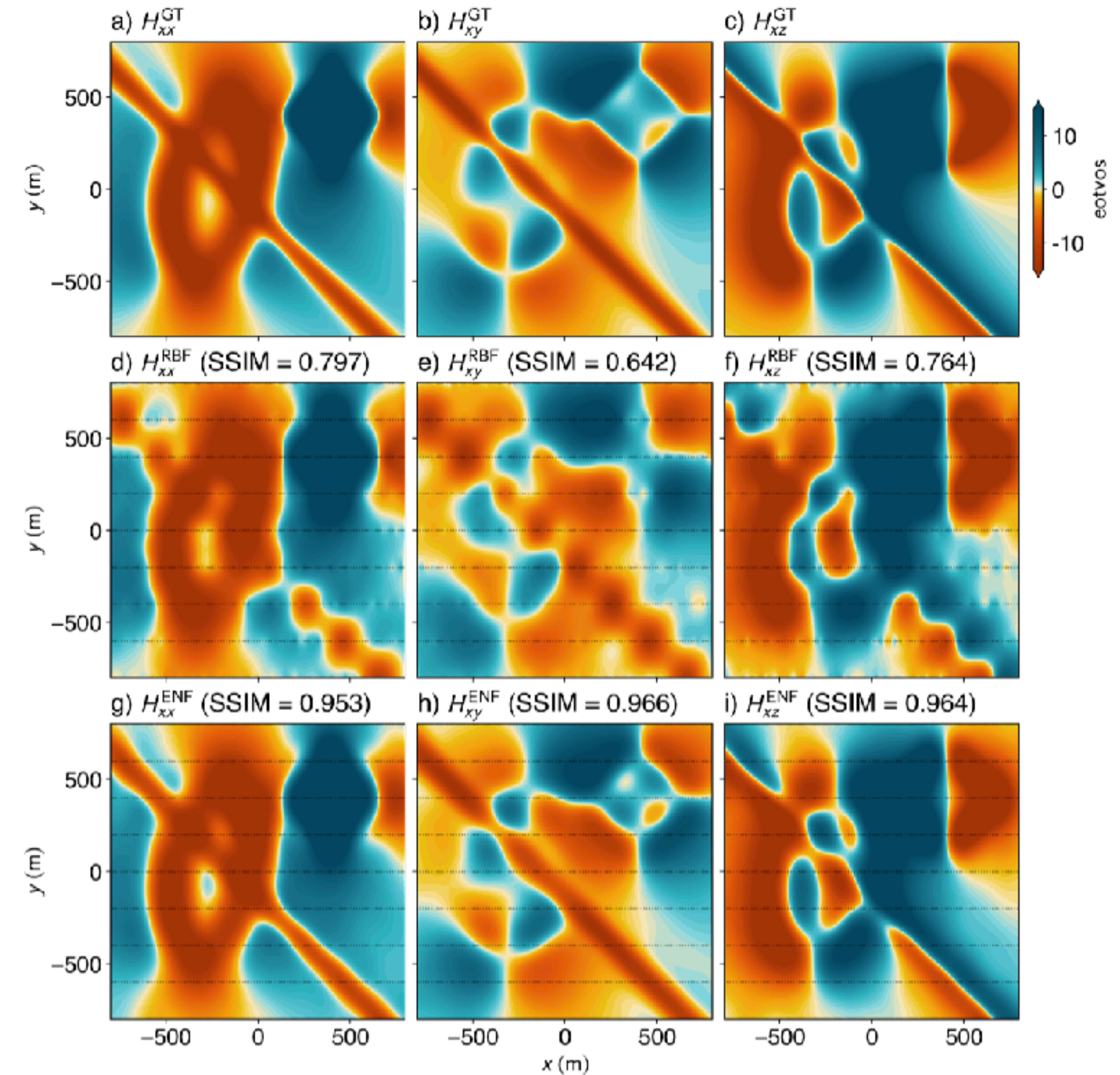
- Pros: physically more consistent, length scales from RFF give better control over feature scale
- Cons: Extensive hyperparameter tuning (data loss and physics loss fighting for influence), choice of points to sample Laplacian loss required.
- Can we somehow include physics in the architecture?
- A partial attempt in TensorWeave: harmonic RFF.
- Why partial? NN after encoding makes output non-harmonic.

Original RFF

$$\mathbf{v}_s = [\sin(2\pi \mathbf{W}_s \mathbf{r}), \cos(2\pi \mathbf{W}_s \mathbf{r})],$$

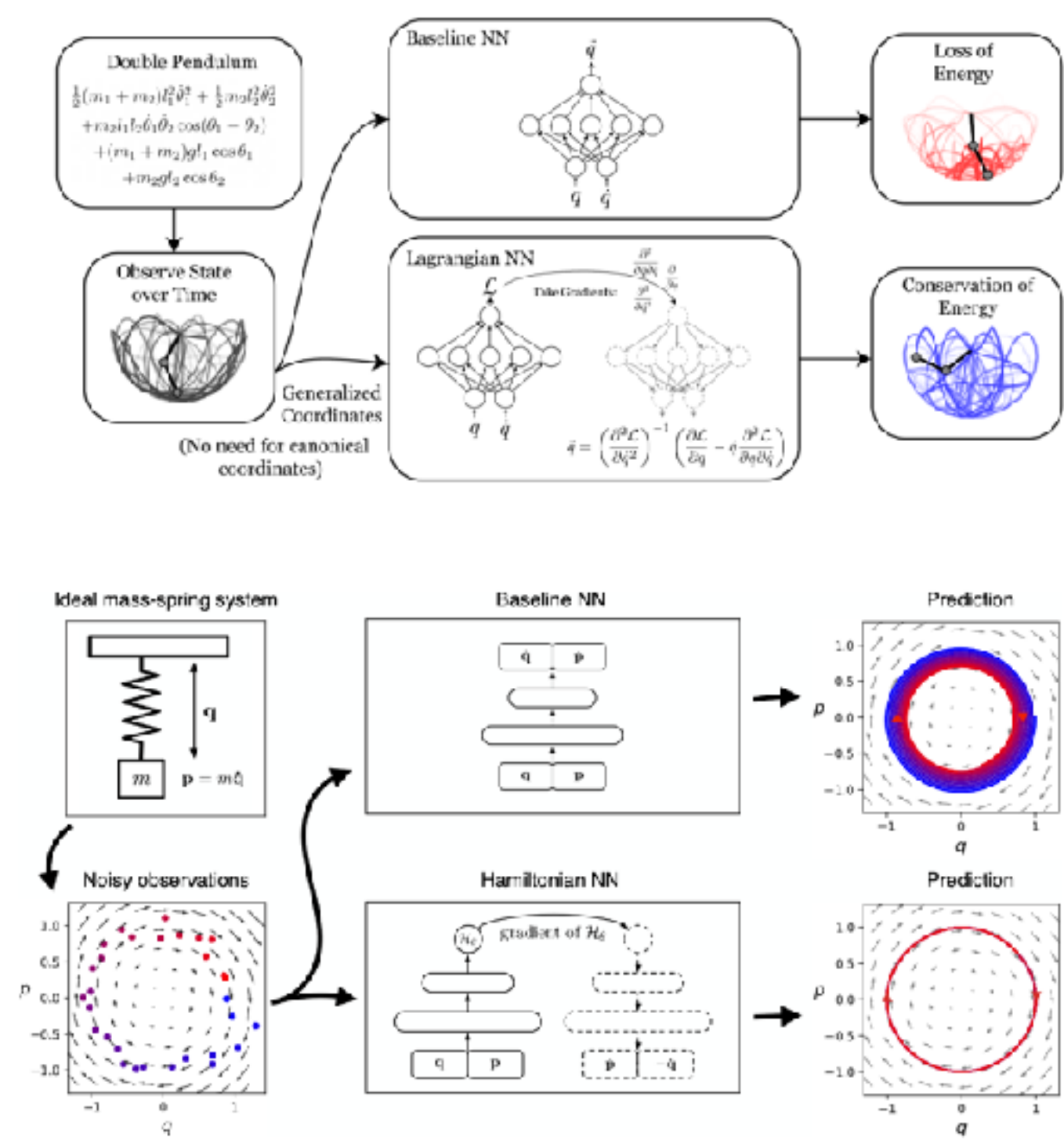
Harmonic RFF

$$\mathbf{v}_s = [\sin(2\pi \mathbf{W}_s \mathbf{r}_{xy}) \odot e^{-\kappa_s z}, \cos(2\pi \mathbf{W}_s \mathbf{r}_{xy}) \odot e^{-\kappa_s z}]$$



Physics in the architecture - avoiding hyper-parameter hell?

- Reparametrize NN to encode physics in the architecture.
- Build invariants into the map (divergence-free, traceless, symplectic), not into the loss with weights.
- Physics-informed = PDE/residuals in the loss (multi-objective hell).
- Physics-in-architecture = the network only outputs admissible fields.



LAGRANGIAN NEURAL NETWORKS

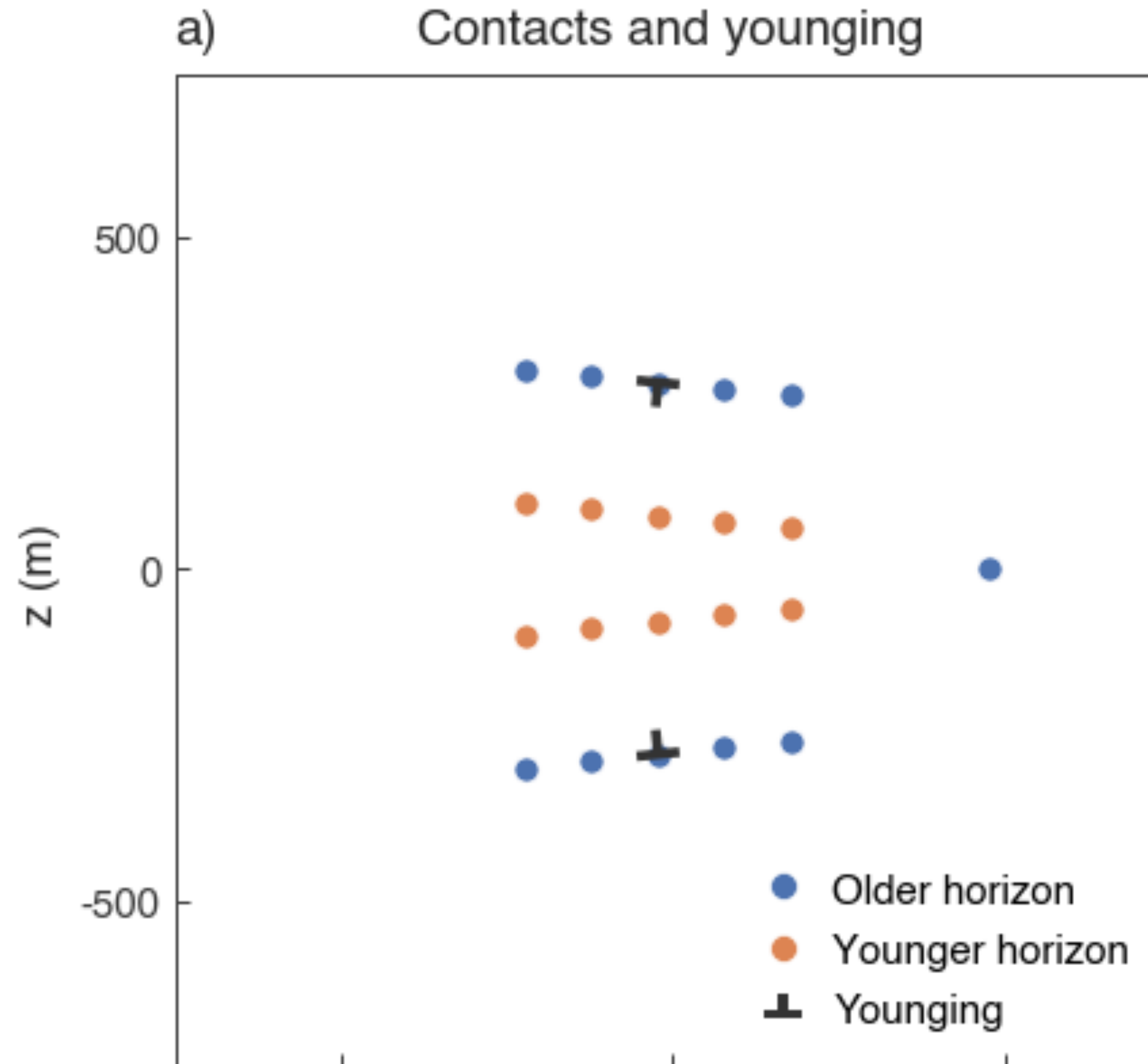
Miles Cranmer Princeton University mcranmer@princeton.edu	Sam Greydanus Oregon State University greydanus.17@gmail.com	Stephan Hoyer Google Research shoyer@google.com	Peter Battaglia DeepMind peterbattaglia@google.com
David Spergel* Flatiron Institute davidspergel@flatironinstitute.org	Shirley Ho† Flatiron Institute shirleyho@flatironinstitute.org		

Hamiltonian Neural Networks

Sam Greydanus Google Brain sgrey@google.com	Misko Dzamba PetCube mouse9911@gmail.com	Jason Yosinski Uber AI Labs yosinski@uber.com
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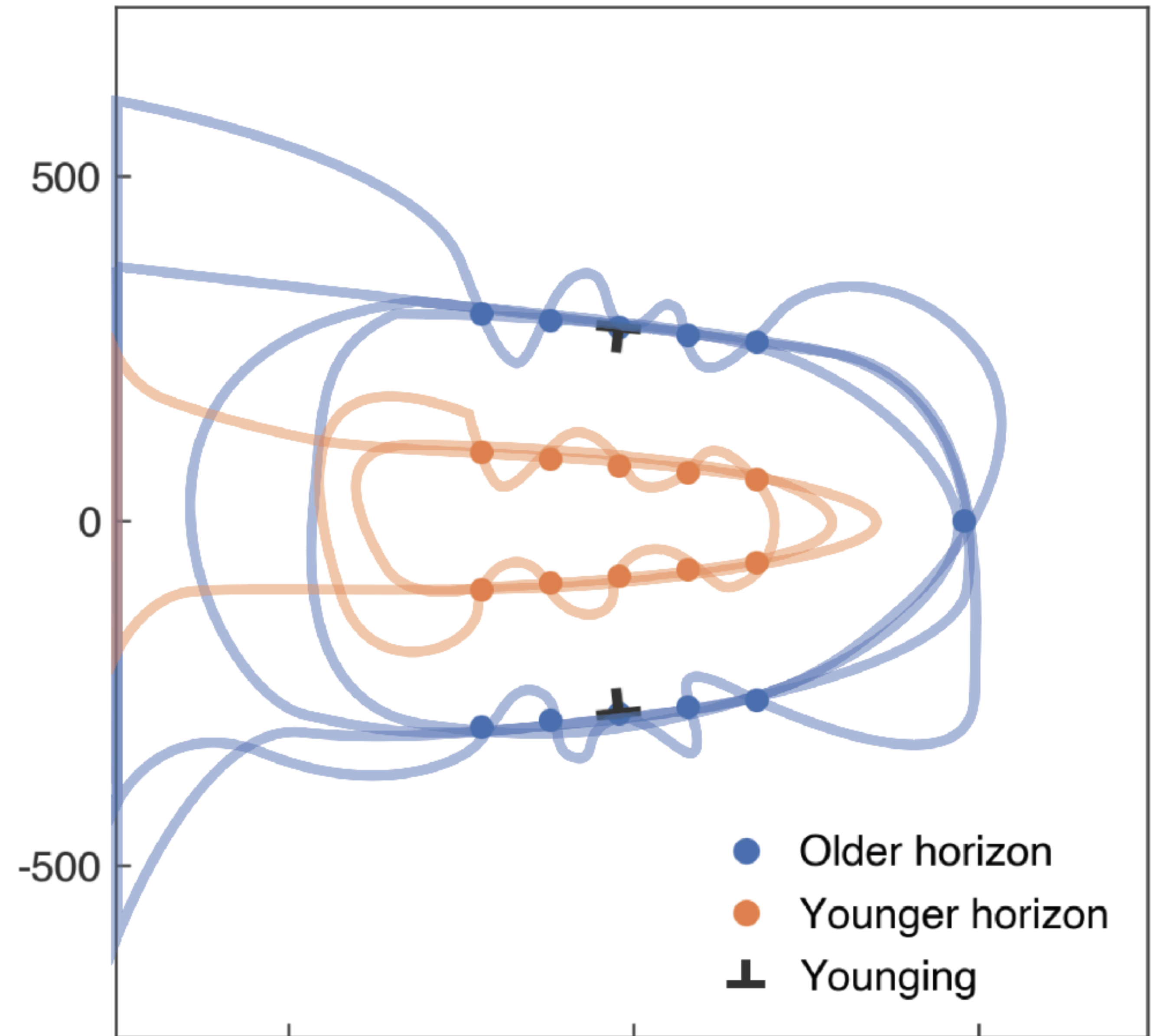
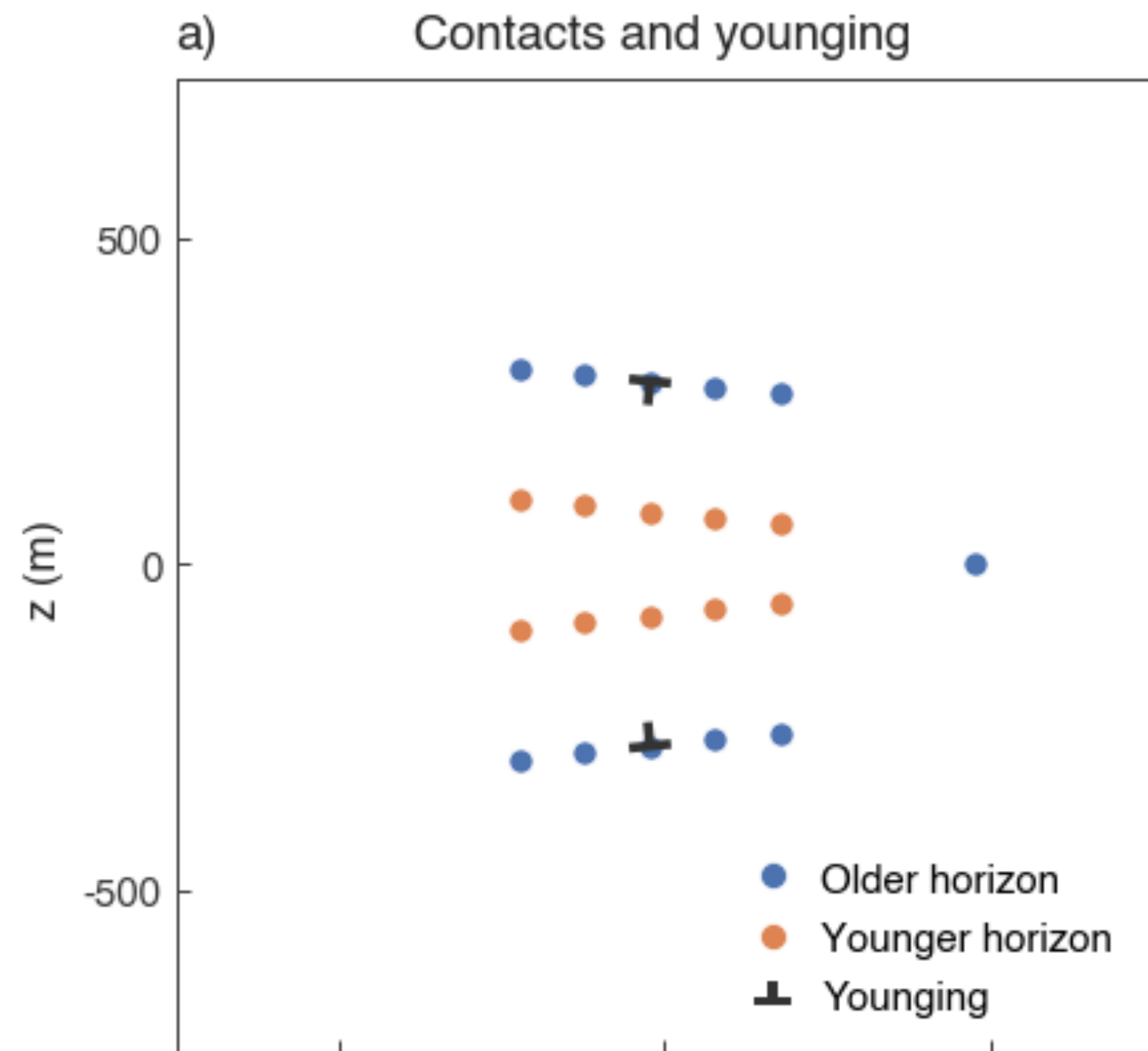
Topology in the architecture

How would you connect these dots?



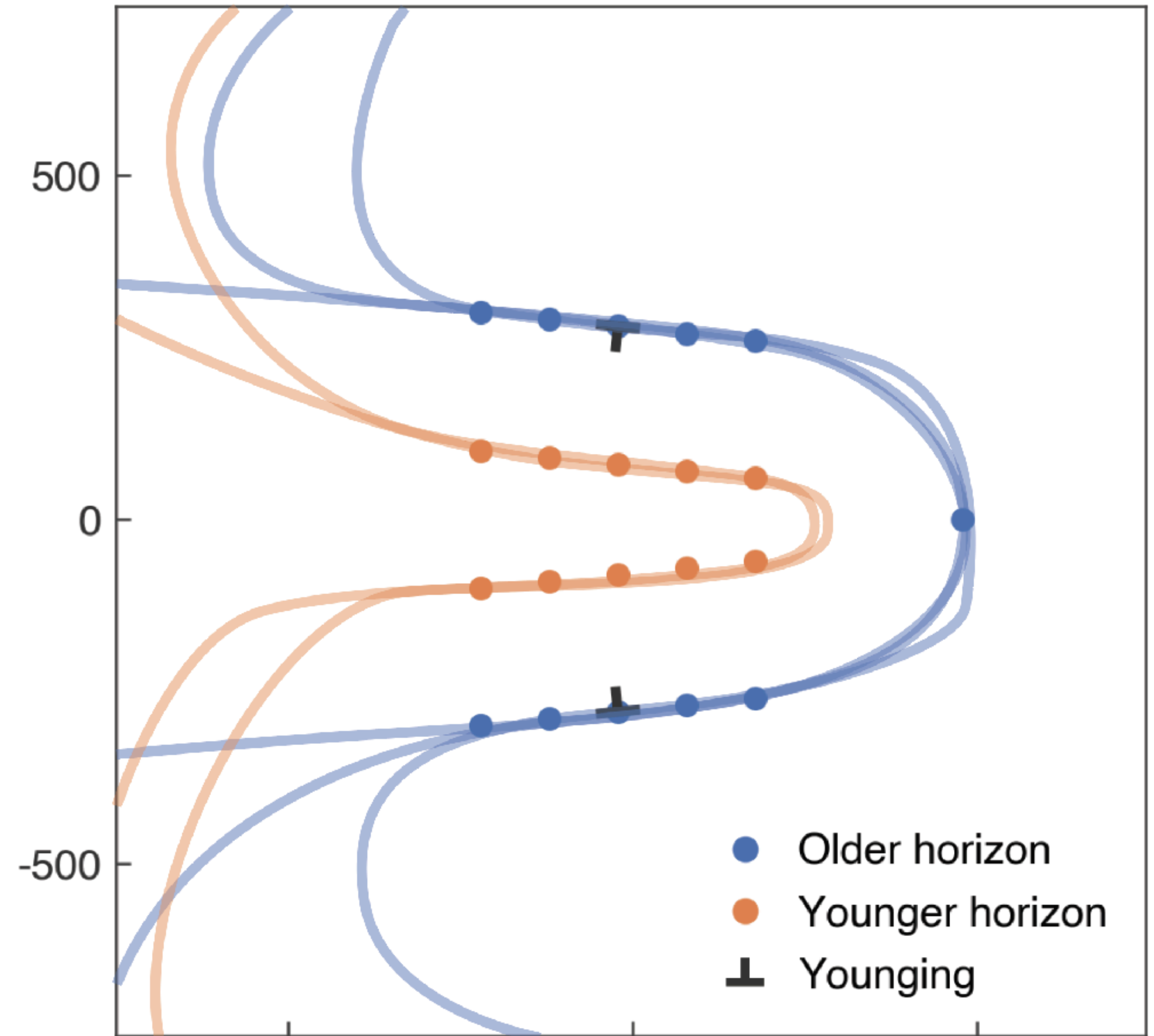
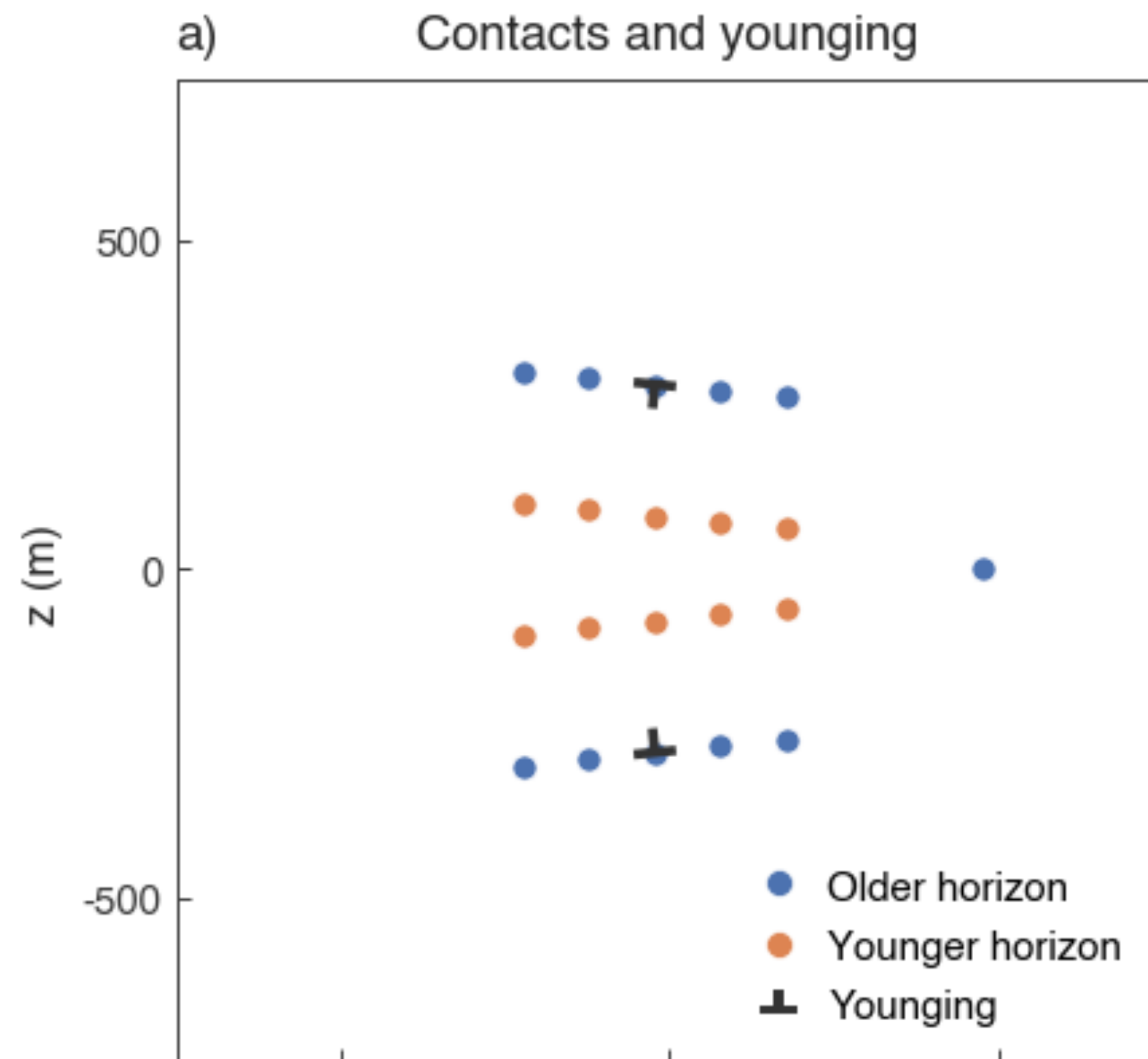
Topology in the architecture

How would you connect these dots?



Topology in the architecture

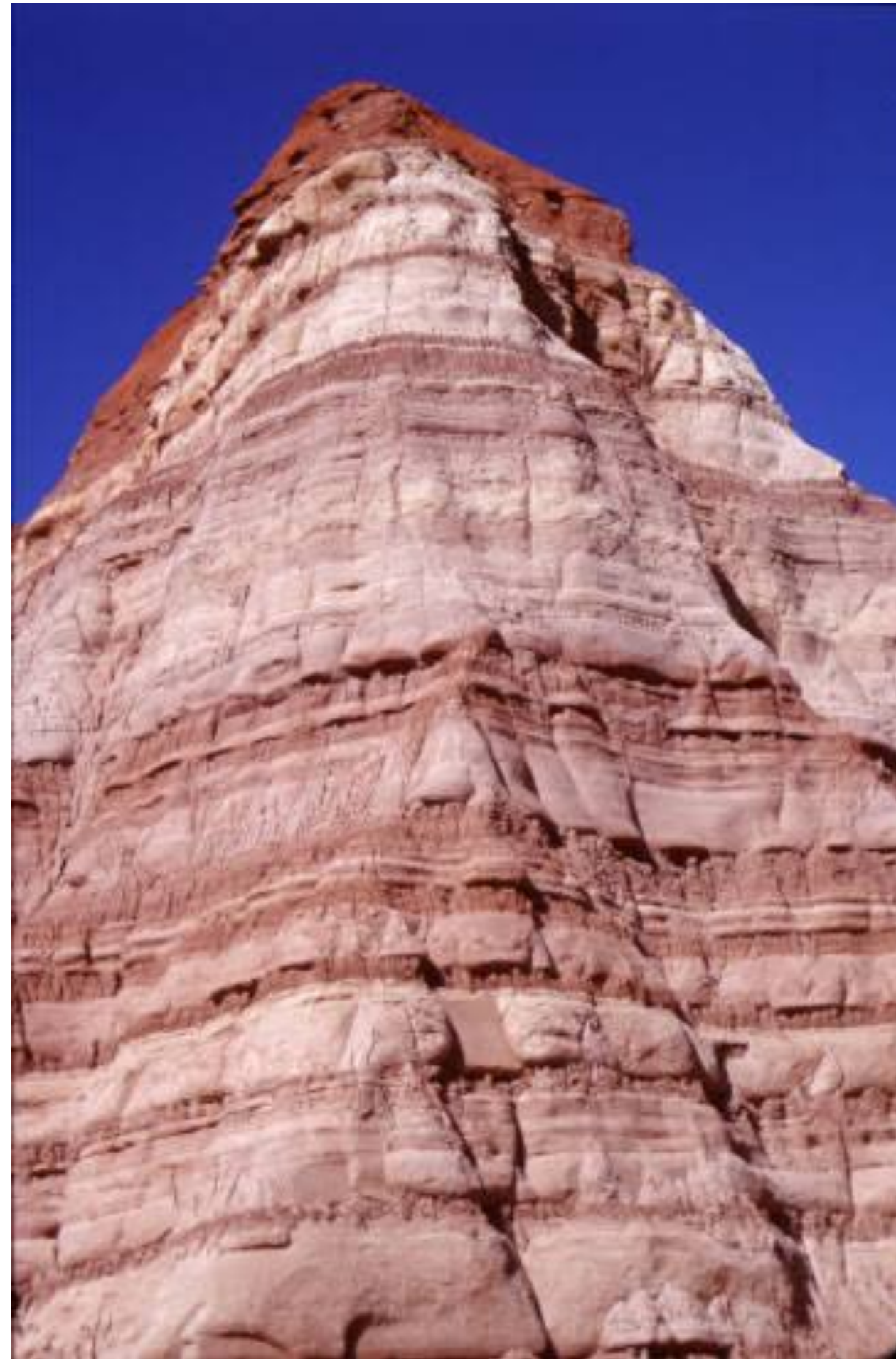
How would you connect these dots if they cannot close in on themselves?



Topology in the architecture

- Physics alone not enough.
- E.g., in geology, rock layers are initially flat -> get folded into complex geometry.
- Rock layers cannot intersect - topology is not altered by slow continuous folding.
- Instead of interpolation, invert deformation: Learnable Diffeomorphic Interpolators (LDI).
- By making the network learn a deformation as a diffeomorphism, final geometry cannot break reference topology - topology in architecture.

Reference

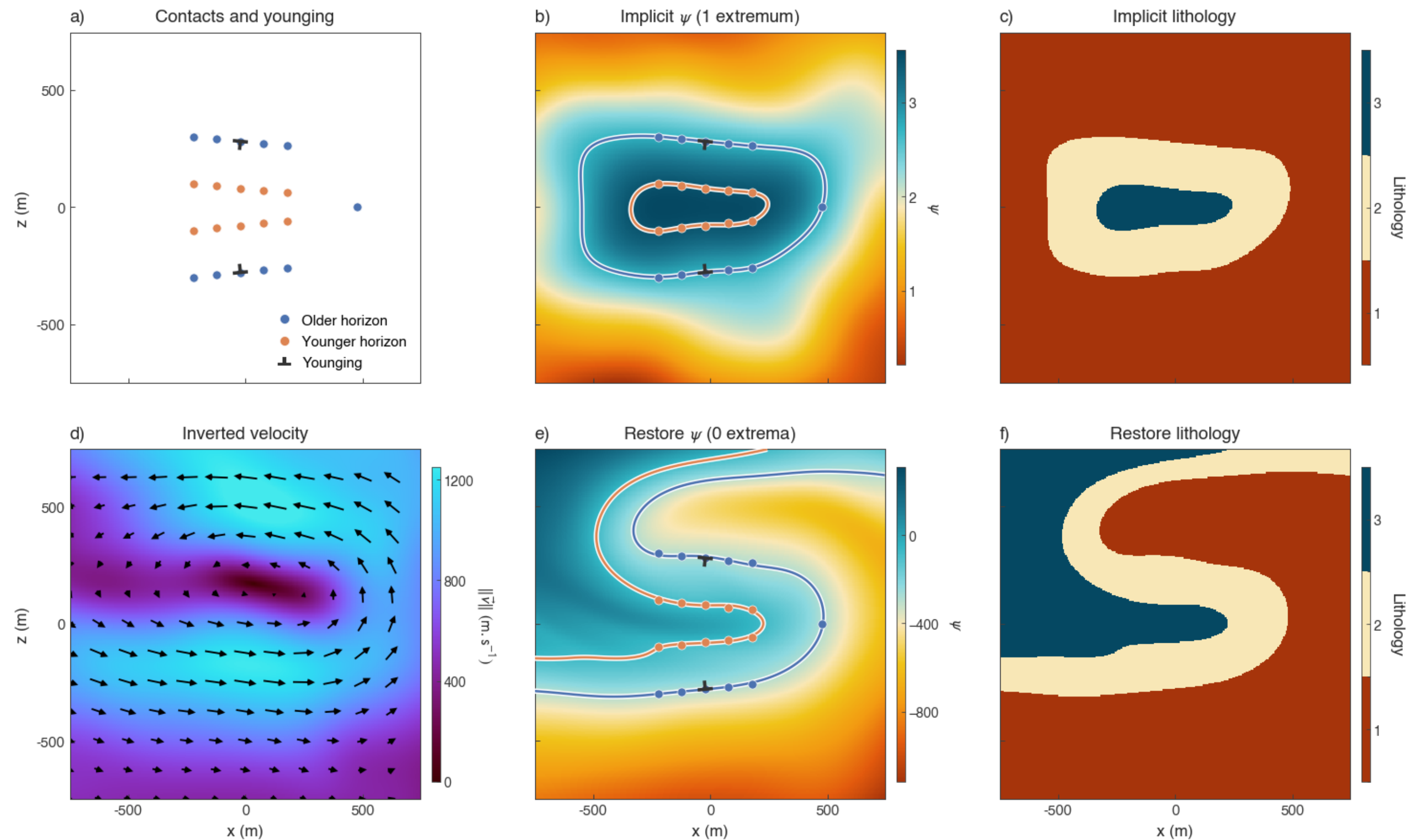


Turns into



Topology in the architecture - Results

- Recumbent fold dataset: implicit interpolation gives topologically wrong solution.
- Learnable deformation preserves initial state topology.
- No extra loss terms - architectural reparametrization.
- Additional physics in the underlying velocity field (like volume-preservation).



Key takeaway: Making the NN emulate physics within the architecture and reparametrizing the process to preserve other properties, helps learning components focus on data fitting!

Conclusions

“...only a fraction of the available geological and geophysical information is used [in geological models]”

- Wellmann & Caumon, 2018



Curlew 1.0: Spatio-temporal implicit geological modelling with neural fields in python

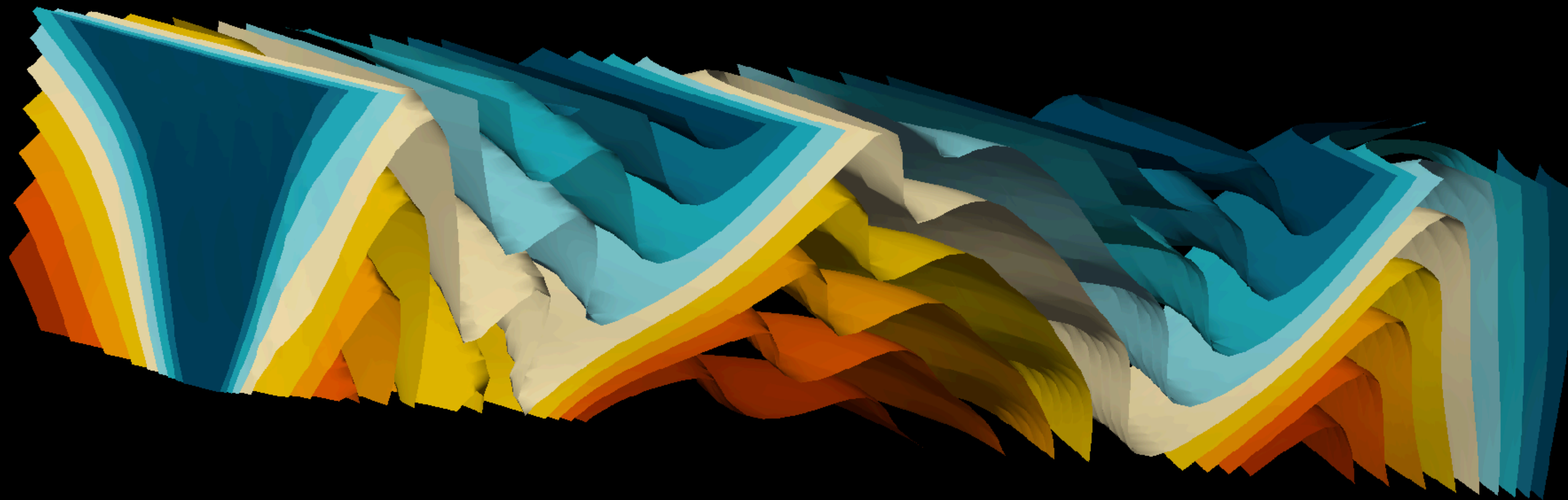
Akshay V. Kamath^{1,*}, Samuel T. Thiele¹, Marie Moulard², Lachlan Grose³, Raimon Tolosana-Delgado¹, Michael J. Hillier⁴, Florian Wellmann^{5,6}, and Richard Gloaguen¹

Physically and topologically constrained interpolation via learnable diffeomorphic interpolators

Akshay V. Kamath^{*1,3}, Samuel T. Thiele¹, Michael J. Hillier², David Nathan³, Lachlan Grose⁴, and Richard Gloaguen¹

(Coming soon!)

A quick break — then code!



<https://tinyurl.com/HZDRAI-GIT>

<https://tinyurl.com/HZDRAI-1>

<https://tinyurl.com/HZDRAI-2A>

<https://tinyurl.com/HZDRAI-2B>

Discussion

Ideas and future directions

- > Are there better way to optimize multi-objective losses (like PINNs)
- > Other/better approaches to restrict Neural Network solutions to some physically plausible space?
- > Any other thoughts / advice / questions :-)



Thank you!

Questions?



*curl***ew** on Github! Give
us a star if you like it :-)



Yours truly on Github...
@k4m4th